Rel-Sortability of Time-Unrolled Causal Graphs: Exact Values, Finite-Sample Saturation, and Pre-Registered Real-Data Tests

Anonymous Authors[1]

Abstract

Causal-discovery benchmarks can be solved by sorting variables by marginal statistics (var-, R^2- and rel-sortability). Time-unrolled DAGs were proposed as a remedy, with a sufficient condition for rel-sortability to tend to $1/2$. We give the exact rel-sortability of the time-unrolled DAG for every summary graph, every horizon T and every edge, with an explicit $O(1/T)$ rate. For the finite DAG the earlier condition is also necessary: every other summary graph, for instance any river network, has a finite-DAG value strictly above $1/2$. This excess is caused by the fixed start at $t = 1$: with a stationary start the value is exact for every T and depends only on the source strongly connected components. Without self-loops the exact edge value survives, but the closed form and the converse fail (counterexamples). For Erdős–Rényi summary graphs we derive the limiting values, which peak near the percolation threshold at 0.754 (finite) and 0.665 (stationary); this derivation is a sketch whose concentration step is not proved, and it agrees with exact computation to 3–4 digits. Block-mean aggregation and point subsampling leave the relatives unchanged but move var- and R^2-sortability to different limits. Raw correlation tests are classically inconsistent under autocorrelation; we add an effective sample size for the relatives count, a non-asymptotic recovery bound, saturation by a latent driver at $n \approx z_c^2/\ell^2$, and conditions for a proxy to repair it. The graph results and the algebraic core of the finite-sample results are machine-checked in Lean 4 (133 theorems). In pre-registered real-data studies a river benchmark fails as the theory predicts, and a physical testbed gives a partial positive result, including out-of-sample predictions of detection rates. A fourth pre-registration (E1) shows that the theorem's edge values beat eight naive predictors on a light-tunnel series, mostly through predicted ties and only for linearly dependent relatives (on the published graph one naive predictor is significantly better on strictly ordered edges). A lagged-relatives estimator (E2) fails its pre-registered graded criterion, as P9 did. In short, the paper completes the unrolled-graph analysis of the earlier work and adds a finite-sample and real-data layer.

1. Introduction

Simulated benchmarks drive much of causal discovery, and several of their regularities can be exploited without any causal reasoning. Marginal variances tend to grow along the causal order (*var-sortability*; Reisach et al., 2021), so a method that sorts by variance does well on data from common generators, and it explains the strong benchmark performance of some continuous-optimisation methods, a family that includes NOTEARS (Zheng et al., 2018). The coefficient of determination R^2 has the same property and is scale invariant (Reisach et al., 2023); internally standardised structural models are not var-sortable by construction and, empirically, mostly not R^2-sortable (Ormaniec et al., 2025); and Herman et al. (2025) relate the R^2 pattern to colliders and confounders and propose a generator that avoids it. Most recently, Reisach et al. (2026) defined *rel-sortability*, which orders a node by the number of its *relatives* (nodes joined to it by an open path, marginally dependent on it under faithfulness). Relatives are a property of the graph alone, so the criterion cannot be removed by reweighting. For random DAGs it is high; for *time-unrolled* DAGs built from a summary graph with self-loops, Reisach et al. (2026, Thm. 4) show that it tends to $1/2$ if every weakly connected component of the summary graph is strongly connected, and they propose such graphs as benchmarks. Only sufficiency is shown in the arXiv version; they do not analyse the value beyond this case.

Three questions remain open, and they matter for real time series, whose summary graphs are often feed-forward

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

(rivers, supply chains, signalling cascades) and whose observations are aggregated or subsampled: (Q1) what is the rel-sortability of a time-unrolled DAG when the summary graph is not strongly connected; (Q2) does it depend on how the series is started and observed; and (Q3) can a relatives count be estimated from one finite, autocorrelated record, so that a method could actually exploit it?

Contributions. **(1)** The exact rel-sortability of every edge of the time-unrolled DAG G_T, for every summary graph with self-loops and every T (Theorem 1), its limit $\frac{1}{2} + |E_\times|/(2|E_\mathrm{S}|)$ with $E_\times$ the edges between strongly connected components, and an explicit $O(1/T)$ rate (Theorem 2). The condition of Reisach et al. (2026, Thm. 4) is also necessary (Corollary 3). Without self-loops the exact edge value still holds, but the closed form and the converse are false (Section 3.1). The excess above $1/2$ for, e.g., chains and out-trees is a boundary effect of the deterministic start at $t = 1$ (Proposition 4). **(2)** A stationary version (latent burn-in of $p - 1$ slices or an infinite past) whose value is exact for every T and is fixed by the *source* strongly connected components (Theorem 5); river networks exceed $1/2$ iff they contain a confluence (Corollary 6). **(3)** Random summary graphs (Section 3.2): limits $L(c)$, $S(c)$ for directed Erdős–Rényi graphs with mean degree c (a sketch with numerical support: the concentration step and the second fixed point are not proved), exact in-forest identities, and an elementary reading of s_ρ as the accuracy of the ideal count-sort-orient rule. **(4)** Under aggregation the relatives are unchanged, whereas var- and R^2-sortability converge to those of a long-run structural model under block means (a model due to Gong et al., 2017; we add the sortability consequences and a rate) and stay at the snapshot values under subsampling (Section 4). **(5)** Finite-sample theory (F1–F3, Section 5). Inconsistency of the nominal-n test under autocorrelation is classical (Bartlett, 1935; Clifford et al., 1989; Roy, 1989; Afyouni et al., 2019); we add an effective sample size and a non-asymptotic recovery bound for the relatives count, saturation by a latent driver at $n \approx z_c^2/\ell^2$, and the conditions for a proxy to repair it, in line with known results on unreliable covariates (Westfall & Yarkoni, 2016). **(6)** Lean 4 proofs of all graph results, the loop-free edge formula and the deterministic core of F1–F3 (133 theorems, standard axioms only; Section 6); to our knowledge the first machine-checked development of sortability results, while Lean libraries for other parts of causal inference exist (Tan & Syrgkanis, 2026). **(7)** Four pre-registered real-data studies (Section 7), to our knowledge the first pre-registered tests of rel-sortability on real data (var- and R^2-sortability were measured on river and chamber data by Lohse & Wahl, 2025): CausalRivers (Stein et al., 2025) fails as F2/F3 predict, and the Causal Chambers (Gamella et al., 2025) give a partial positive result, with E1 and E2 as described in the abstract. We report every failed pre-registered test (Table 2).

The graph results of contributions (1)–(2) are short consequences of a walk-length lemma and of Reisach et al. (2026, Lemma 1). Their value is that they determine the value for every summary graph and horizon, are machine-checked, and change what a benchmark designer should conclude: much of the rel-sortability of time-unrolled benchmarks is an artefact of how the simulation is started. The random-graph results are a sketch plus numerics rather than a proof. The finite-sample results use textbook tools; their value is that they predicted, quantitatively and in one case out of sample, the outcome of real-data tests. A referee should read the whole as a completion of the unrolled-graph analysis of Reisach et al. (2026) (their Section 5) plus a finite-sample and real-data layer, not as a new theory of sortability.

Related work. Varsortability (Reisach et al., 2021) and R^2-sortability (Reisach et al., 2023) have been analysed for standardised SCMs (Ormaniec et al., 2025), for unit-free generators with a time-series extension (Herman et al., 2025), and, for time series, by Lohse & Wahl (2025), who adapt both criteria to summary graphs, note that cyclic pairs push the signal towards $1/2$, and measure them on river-flow and Causal Chambers data; they give no theorem and do not study relatives or observation schemes. Reisach et al. (2026) introduce rel-sortability and the time-unrolled DAG, and prove that strong connectivity of every weak component of the summary graph is sufficient for the limit $1/2$. We determine the value exactly for every summary graph, horizon and start, and show that the condition is necessary for the finite DAG. Block means of a VAR converge to a static linear SEM with coefficients $A_{ji}/(1 - A_{jj})$ (Gong et al., 2017); that work and Gong et al. (2015) study identifiability of the causal-frequency model from aggregated or subsampled data, not sortability. The inconsistency of a Pearson test with the nominal sample size under autocorrelation is classical (Bartlett, 1935; Clifford et al., 1989; Roy, 1989) and is used in practice (Runge et al., 2019; Afyouni et al., 2019); we add an effective sample size for the relatives count with a non-asymptotic recovery bound, and quantify how a latent driver or an imperfect proxy defeats it, in line with known residual-confounding results for unreliable covariates (Westfall & Yarkoni, 2016). Formal libraries for causal inference exist (Tan & Syrgkanis, 2026); we machine-check the sortability results and the deterministic core of the finite-sample theory. Identifiability from subsampled and aggregated series is a separate question from sortability, and time-series discovery methods such as PCMCI (Runge et al., 2019) test conditional independences, whereas our finite-sample results concern only the marginal tests that define relatives.

2. Setting

For a directed graph, $\mathrm{anc}(y)$ and $\mathrm{desc}(y)$ contain y; the relatives of y are $\mathrm{rel}(y) = \mathrm{desc}(\mathrm{anc}(y)) = \{z : \mathrm{anc}(y) \cap \mathrm{anc}(z) \neq \emptyset\}$ and $\rho(y) = |\mathrm{rel}(y)|$. For an edge (x, y), $s_\rho(x, y) = 1, \frac{1}{2}, 0$ if $\rho(x) <, =, > \rho(y)$, and $s_\rho(G)$ is the mean over edges (Reisach et al., 2026). A value $1/2$ means the criterion carries no orientation information.

A *summary graph* $G_\mathrm{S} = (V, E_\mathrm{S})$ has $p \geq 2$ nodes and a self-loop at every node. The *time-unrolled DAG* G_T has nodes $v(t)$, $t \in \{1, \ldots, T\}$, and an edge $(i(t), j(t{+}1))$ for each $(i, j) \in E_\mathrm{S}$ and $t < T$, so every summary edge has $T - 1$ copies. $d(u, v)$ is the shortest-path length in G_S (self-loops ignored; $d(v, v) = 0$; ∞ if unreachable), and anc_S, rel_S are taken in G_S. SCC means strongly connected component. $E_\times \subseteq E_\mathrm{S}$ is the set of edges between different SCCs and $E_\circ = E_\mathrm{S} \setminus E_\times$. A *source* SCC has no incoming edge from another SCC, and $\mathrm{Src}(v)$ is the set of source SCCs that reach v.

3. Exact Rel-Sortability of Time-Unrolled DAGs

Theorem 1 (every edge, every T)**.** *For $(i, j) \in E_\mathrm{S}$ and $1 \leq t < T$, the edge $(i(t), j(t{+}1))$ of G_T has $s_\rho = 1$ if some $u \in V$ satisfies $d(u, j) \leq t \leq d(u, i)$, and $s_\rho = 1/2$ otherwise. The value does not depend on T.*

Lean: `edge_score_dist`, `edge_score`, `strictCond_iff_dist`.

Sketch. Every edge of G_T advances time by one, so $u(s)$ is an ancestor of $v(t)$ iff $d(u, v) \leq t - s$ (pad a shortest path with self-loops). Hence the roots of $v(t)$ are $\{u(1) : d(u, v) \leq t - 1\}$. Since $i(t)$ is a parent of $j(t{+}1)$, $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$, and the inclusion is strict iff $j(t{+}1)$ has a root that $i(t)$ lacks (Reisach et al., 2026, Thm. 1, Lemma 1). Full proof in Appendix A. (The Lean proof characterises rel in G_T directly and does not use their Lemma 1.)

Theorem 2 (exact limit and rate)**.** *For every $T \geq 2$, with $L(G_\mathrm{S}) = \frac{1}{2} + \frac{|E_\times|}{2|E_\mathrm{S}|}$,*

$$0 \leq s_\rho(G_T) - L(G_\mathrm{S}) \leq \frac{\min(p{-}1, T{-}1)\,|E_\circ|}{2(T-1)|E_\mathrm{S}|} \leq \frac{p-1}{2(T-1)},$$

so $\lim_{T\to\infty} s_\rho(G_T) = L(G_\mathrm{S})$.

Lean: `sRho_unrolled_exact`, `sRho_unrolled_bounds`, `tendsto_sRho_GT`.

Sketch. For a crossing edge, j does not reach i; take $u = j$ in Theorem 1: all $T - 1$ copies score 1. For an edge inside an SCC, i and j are reached by the same nodes, which forces $t \leq d(u, i) \leq p - 1$, so at most $p - 1$ copies score 1.

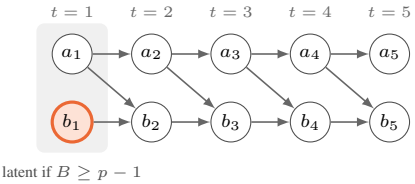


Figure 1. Summary chain $a \to b$ unrolled to $T = 5$. Every a_t has the relatives $\{a_s\} \cup \{b_s : s \geq 2\}$ ($2T - 1$ nodes); every b_t with $t \geq 2$ also has b_1 ($2T$ nodes). So each edge $a_t \to b_{t+1}$ is strictly sorted (finite limit $L = 2/3$), but only because of the root b_1 at the fixed start. With a latent burn-in slice, both counts are $2T$ and $s_\rho^W = 1/2$.

Corollary 3 (necessary and sufficient condition)**.** $\lim_T s_\rho(G_T) = 1/2$ *iff every weakly connected component of G_S is strongly connected. Otherwise $s_\rho(G_T) \geq \frac{1}{2} + \frac{1}{2|E_\mathrm{S}|}$ for every T. The "if" half is Reisach et al. (2026, Thm. 4). If G_S is acyclic apart from self-loops, with m other edges, $L = \frac{1}{2} + \frac{m}{2(m+p)}$ whatever its shape (chain, in-tree, out-tree, collider).*

Lean: `tendsto_half_iff`, `crossEdges_eq_empty_tfae`, `sRho_unrolled_ge_of_cross`, `reisach_theorem4`, `limitValue_dag`.

Proposition 4 (the excess is a boundary effect)**.** *For $t \geq p$, $\rho(v(t)) = T\,|\mathrm{rel}_S(v)| - \sum_{w \in \mathrm{rel}_S(v)} d(\mathrm{anc}_S(v), w)$.*

Lean: `rho_unrolled_large_t`, `rho_unrolled_eq`.

So for a crossing edge with $\mathrm{rel}_S(i) = \mathrm{rel}_S(j)$ (root to child in an out-tree, a chain), the strict order rests on $O(1)$ extra relatives out of $\Theta(T)$, all located in the first p slices. Figure 1 shows the smallest case. A sample-based relatives count on a long record cannot see such a difference, which motivates the stationary version.

Theorem 5 (stationary start)**.** *Observe the window of the last $T \geq 2$ slices of G_{B+T} with a latent burn-in $B \geq p - 1$ (or of the infinite-past graph on $V \times \mathbb{Z}$), with relatives computed in the full graph. Then every window node has $\rho_W(v(t)) = T|\mathrm{rel}_S(v)|$, and for every T*

$$s_\rho^W = \tfrac{1}{2} + \frac{|E_\uparrow|}{2|E_\mathrm{S}|},\ E_\uparrow = \{(i, j) \in E_\mathrm{S} : \mathrm{Src}(i) \neq \mathrm{Src}(j)\}.$$

Lean: `rhoBurn_eq`, `sBurn_eq`, `rhoZ_eq`, `sZ_eq`, `relSet_eq_iff_src`.

Sketch. After $p{-}1$ slices every finite distance is absorbed, so $w(s) \in \mathrm{rel}(v(t))$ iff $\mathrm{anc}_S(v) \cap \mathrm{anc}_S(w) \neq \emptyset$. For an edge (i, j), $\mathrm{rel}_S(i) \subseteq \mathrm{rel}_S(j)$ with equality iff $\mathrm{Src}(i) = \mathrm{Src}(j)$.

Corollary 6 (stationary condition; graph classes)**.** $s_\rho^W = 1/2$ *iff every weakly connected component of G_S has exactly one source SCC. Strongly connected graphs, out-forests,*

Table 1. Rel-sortability of 3-node summary graphs (with self-loops, $|E_S| = 5$). Finite: $\lim_T s_\rho(G_T)$ (Theorem 2); stationary: s_ρ^W (Theorem 5). Lean: `chain3_limit`, `chain3_stat`, `fork_limit`, `fork_stat`, `collider_limit`, `collider_stat`; 3-cycle by `reisach_theorem4`.

summary graph	finite G_T (limit)	stationary
chain $a \to b \to c$	0.7	0.5
fork $a \leftarrow b \to c$	0.7	0.5
collider $a \to c \leftarrow b$	0.7	0.7
3-cycle (6 edges)	0.5	0.5

chains and forks give $1/2$. In an in-forest (a river network), $s_\rho^W = \frac{1}{2} + \#\{\text{edges into confluences}\}/(2|E_S|)$, which exceeds $1/2$ iff there is a confluence.

Lean: `sBurn_eq_half_iff`, `sZ_eq_half_iff`, `stat_half_of_outForest`, `statValue_inForest`, `statValue_inForest_gt_iff`, `collider_stat`, `fork_stat`.

Table 1 shows the two versions. In the finite generator every summary DAG with the same m/p has the same value, so a benchmark built this way rewards relatives-sorting on chains and forks, where the stationary process gives no signal; for long stationary records the stationary value is the relevant one and is what our real-data predictions use. All statements of this section were checked by exhaustive enumeration ($p \le 4$ and 10^4 random graphs with $p \le 7$; 46,724 cases, 0 failures; Appendix B).

3.1. Without self-loops

Without self-loops an ancestor needs a walk of *exactly* the right length, and nodes with no incoming edge ("fresh") inject a new root at every slice. The exact edge value survives: for any summary graph every edge of G_T scores $1/2$ or 1, independently of T, by an explicit criterion (Theorem 22, Appendix K), so the finite-T value is exact. The limit exists and is rational (eventual periodicity; paper proof only), and Reisach et al. (2026, Thm. 4) still holds. The closed form $L = \frac{1}{2} + |E_\times|/(2|E_S|)$ and the converse of Corollary 3 do not: for the single edge $0 \to 1$ without loops $s_\rho(G_T) = 1/2$ for every T; for the 2-cycle $\{0, 1\}$ plus the edge $0 \to 2$, all without loops, $L = 1/2$ although $(0, 2)$ is crossing (with a loop at 2, $L = 5/8$). So the loop assumption is necessary for the closed form and the converse.

Lean: `edge_score_NL`, `sRho_unrolled_eq_NL`, `sRho_unrolled_ge_half_NL`, `edgeNoLoops_sRho`; limit, periodicity and the stationary version without loops are paper-only or enumerated.

3.2. Random summary graphs

Let G_S be the directed Erdős–Rényi graph $G(p, c/p)$ plus a self-loop at every node, and x the largest root of $x = 1 - e^{-cx}$ ($x = 0$ for $c \le 1$). By Theorems 2 and 5 only $|E_\times|$ and $|E_\uparrow|$ matter.

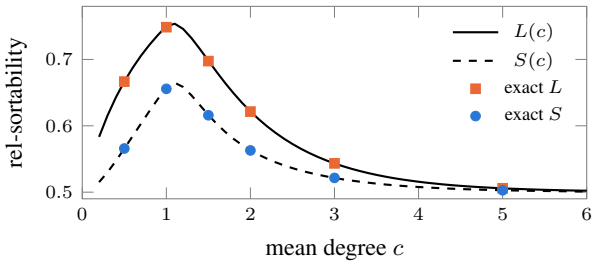


Figure 2. Proposition 7 (lines) against exact computation on seeded graphs ($p = 2 \cdot 10^4$ to 10^5; points, means over seeds). The standard deviations over seeds are at most 0.0022 (Appendix L).

Proposition 7 (ER limits; sketch)**.** *In probability as $p \to \infty$, $L(c) = \frac{1}{2} + \frac{c(1-x^2)}{2(1+c)}$ and $S(c) = \frac{1}{2} + \frac{c\,F_\uparrow}{2(1+c)}$ with $F_\uparrow = (1-x)(1-e^{-c}) + x(1-x)\big(1 - e^{-c(1-x)d}\big)$, where d is the smallest fixed point in $[0, 1]$ of $d = 1 + e^{-c} - e^{-c(1-x)d}$ ($d = 1$ for $c \le 1$).*

Lean: not machine-checked (a local-limit sketch, Appendix L: the concentration step is not proved and the fixed point d is derived heuristically; the formulas agree with exact computation to 3–4 digits).

The first formula says that the fraction of non-loop edges inside an SCC tends to x^2 (giant strongly connected component, Dorogovtsev et al., 2001). $L(1) = 3/4$, L peaks at 0.754 near $c = 1.07$ and then decays like e^{-c}; S peaks at 0.665 near $c = 1.09$ (Figure 2). Random summary graphs are thus rel-sortable only near or below criticality, as the giant SCC removes the signal. For in-forests (rivers; mean parent count μ, fraction π_1 with exactly one parent) the identities are exact: $L \to \frac{1}{2} + \mu/(2(1+\mu))$ is shape-independent while $S \to \frac{1}{2} + (\mu - \pi_1)/(2(1+\mu))$ counts pass-through nodes (Appendix L).

What the value means. Show an edge in random order with the two relatives counts; the rule "orient towards the larger count, flip a coin on ties" has accuracy exactly s_ρ, never errs on an edge it decides, and no pairwise rule on the two counts does better (Reisach et al.'s Theorem 1 plus three lines, Appendix M). So $s_\rho = 0.7$ caps such a method at 70% orientation accuracy: precision 1 on decided edges and coverage $2(s_\rho - \frac{1}{2})$. This is elementary. Rules that also see node degrees can exceed it (Appendix M).

Lean: `accRule_sortRule`, `accRule_le_sRho`, `edgeAcc_sortRule_strict`, `rho_not_gt_of_edge`.

4. Aggregation versus Subsampling

Let X be a stable linear SVAR with lags $\tau \ge 1$, independent innovations with diagonal covariance Σ, and summary graph G_S; $\Gamma(h)$ is its autocovariance and $A(1) = \sum_\tau A_\tau$.

Consider block means $\bar{Y}_s = \frac{1}{D}\sum_{u<D} X(sD+u)$ and subsamples $Z_s = X(sD)$.

Proposition 8 (relatives survive any per-node filter)**.** *If $\mathrm{anc}_S(i) \cap \mathrm{anc}_S(j) = \emptyset$ then X_i and X_j are independent at all lags, and so are any per-node, time-invariant (possibly nonlinear) transforms of them, including $\bar{Y}$ and Z for every D. If the ancestor sets intersect, $\Gamma(0)_{ij}$, $\mathrm{Cov}(\bar{Y})_{ij}$ and the long-run covariance are nonzero for Lebesgue-almost every parameter.*

Lean: not machine-checked (needs the SVAR moving-average representation and real-analytic genericity).

So the population relatives of contemporaneous values are rel_S for every observation scheme, and the population relatives sort equals the stationary value of Theorem 5 (on non-self-loop edges), not the finite-G_T value. Var- and R^2-sortability behave differently.

Theorem 9 (block means vs subsampling)**.** *(a) $\|D\,\mathrm{Cov}(\bar{Y}) - S_{LR}\| \le \frac{1}{D}\sum_h |h|\|\Gamma(h)\|$, with $S_{LR} = (I - A(1))^{-1}\Sigma(I - A(1))^{-\top}$. (b) If G_S is acyclic apart from self-loops, S_{LR} is the covariance of a static SEM (this long-run SEM is due to Gong et al., 2017, Eqs. (5), (10)) on G_S with weights $A(1)_{ji}/(1-\lambda_j)$ and noise variances $\sigma_j^2/(1-\lambda_j)^2$, $\lambda_j = A(1)_{jj} < 1$. (c) $\mathrm{Cov}(Z_s) = \Gamma(0)$ for every D. (d) Hence, for large D, block-mean R^2-sortability is 0 on out-stars and 1 on in-stars with ≥ 2 arms, for all weights; under subsampling every contemporaneous criterion keeps its $\Gamma(0)$ value.*

Lean: (a) counting identity `fejer_identity`; (b) `longrun_factor`, `selfloop_sum_lt_one`; the probabilistic steps, the rate and (c)–(d) are not machine-checked.

We claim the rate (a) and the sortability consequences (d), not the long-run model. Block means can thus reverse the R^2 order a snapshot has: on out-stars, subsampled data are R^2-sorted correctly on average (0.84) while block means drift to 0 (Appendix C). Rel-sortability is the only criterion that depends on the graph alone and is invariant to the observation scheme.

5. Finite-Sample Theory

We now ask whether the population relatives can be estimated. X is a zero-mean stationary Gaussian process with summable autocovariances $\gamma_{ab}(h)$, standardised so that $\gamma_{aa}(0) = 1$, observed at n times. Two nodes are declared relatives if the Fisher-z test of the sample correlation $\hat{r}_{ab}$ rejects at level α with Bonferroni over $m = d(d-1)/2$ pairs, i.e. $|\hat{r}_{ab}| > \tanh(z_c/\sqrt{n'-3})$, $z_c = \Phi^{-1}(1-\alpha/2m)$; $\hat{\rho}(v) = 1 + \#\text{declared partners}$. The *nominal* test uses $n' = n$. Let $B_{ab} = \sum_h |\gamma_{aa}(h)\gamma_{bb}(h) + \gamma_{ab}(h)\gamma_{ab}(-h)|$ (for an independent pair, the Bartlett sum $\sum_h \rho_a(h)\rho_b(h)$).

F1: recovery and effective sample size. That the nominal-n test is inconsistent under autocorrelation is classical: effective degrees of freedom of a correlation go back to Bartlett (1935), with spatial and multivariate versions in Clifford et al. (1989) and Roy (1989), and the variance with exactly our summand B_{ab} is in Afyouni et al. (2019) (see also the supplement of Runge et al., 2019). What is new here is the non-asymptotic exact-recovery bound for the relatives set.

Theorem 10 (recovery bound)**.** *For $x, y > 0$ let $\varepsilon_{ab} = \sqrt{2B_{ab}x/n} + \kappa_2 x/n + 2\kappa_1 y/n$, $\varepsilon_d = \sqrt{2D_\gamma y/n} + 4\kappa_1 y/n < 1$ and $\delta_{ab} = (\varepsilon_{ab} + |\rho_{ab}|\varepsilon_d)/(1-\varepsilon_d)$, with autocovariance constants $D_\gamma, \kappa_1, \kappa_2$ defined in Appendix D. If thresholds separate ($\delta_{ab} \le \tau_{ab}$ on non-relatives and $|\rho_{ab}| - \delta_{ab} > \tau_{ab}$ on relatives), the declared set equals the true one, and hence $\hat{s} = s_\rho^{\mathrm{pop}}$, with probability at least $1 - d(d-1)e^{-x} - 4de^{-y}$. For a common threshold $\tau < \rho_{\min}$ the failure probability decays as $\exp(-n\min(\tau, \rho_{\min} - \tau)^2/(2(\sqrt{B} + \rho_{\max}\sqrt{D_\gamma})^2))$, so n/B plays the role of an effective sample size.*

Lean: deterministic core `thm15_det`, `decide_eq_target`, `corr_perturb`, `sOf_cntDecl_congr`; Chernoff and trace algebra `chernoff_choice`, `phi_le`, `lemma12_trace`, `lemma12_le_B`. The Gaussian tail bound itself is not machine-checked.

For i.i.d. snapshots $B = 1 + \rho^2$; under strong persistence B can be in the tens, and so can the loss in effective sample size. The proof combines a Gaussian-chaos tail bound (the Gaussian case of Hsu et al., 2012, Prop. 1; see also Laurent & Massart, 2000), an exact finite-n variance identity for $\hat{r}$, and a deterministic perturbation lemma.

Proposition 11 (the nominal test is inconsistent; asymptotic)**.** *Assume a joint CLT for the sample variances and cross-covariance of the bivariate stationary Gaussian process (the scalar-process CLTs for quadratic forms are surveyed by Ginovyan & Taqqu, 2021; the joint statement we need is an assumption, not a proved step). Then a non-relative pair is declared by the nominal test with probability $\to 2\bar{\Phi}(z_c/\sqrt{B_{ab}})$, which does not vanish. With $n_{\mathrm{eff}} = n(1-\rho^2)^2/W_{ab}$ (W the delta-method Bartlett variance) the false-positive rate tends to α/m.*

Lean: not machine-checked (relies on the assumed joint CLT, which is not in Mathlib).

F2: saturation by a latent driver. Let $X_v = Y_v + b_v C$ with C a stationary unit-variance process independent of the VAR Y, and $\ell_v = b_v^2/\mathrm{Var}X_v$ the driver's share of variance.

Proposition 12 (saturation)**.** *For summary non-relatives, $\rho_{uv} = \mathrm{sgn}(b_u b_v)\sqrt{\ell_u \ell_v} =: \kappa_{uv}$. If all $b_v \neq 0$, every pair is a population relative and the population rel-sortability is exactly $1/2$; $P(\hat{s}_n \neq \frac{1}{2}) \le (d^2+3d)e^{-cn}$ with $c \approx \kappa_{\min}^2/(18(\sqrt{B} + \sqrt{D_\gamma})^2)$. Asymptotically, a pair is detected with probability $> 1/2$ once $n > n_{1/2} = 3 + (z_c/\operatorname{atanh}\kappa)^2 \approx z_c^2/\ell^2$ (nominal test; for the effective*

test, the same bound on n_{eff}).

Lean: `prop21_corr_std`, `sOf_saturated`, `abs_sOf_sub_half_le`, *location of* $n_{1/2}$*:* `ncrit_eq_iff`*; the probability statements are not machine-checked.*

$n_{1/2}$ does not depend on autocorrelation, which only sets the width of the transition. A driver with share $\ell = 0.1$ saturates at $n \approx 10^3$, and one with $\ell = 0.64$ at $n \approx 25$. A relatives count on a long record can recover the summary relatives only while $\max_{\text{non-rel}} |\kappa| < \tau_n < \min_{\text{rel}} |\rho|$, a window that closes at n of order $n_{1/2}$. Autocorrelation (Proposition 11) and a latent driver are separate failure modes; both push $\hat{s}$ towards $1/2$.

F3: conditioning on a proxy. Let $Z = \sqrt{r}\, C + \sqrt{1-r}\, e$ with reliability $r = \text{corr}(Z, C)^2$, and let π_{uv} be the partial correlation given Z.

Proposition 13 (conditioning)**.** *With ρ^Y the driver-free correlation,*

$$\pi_{uv} = \frac{\rho^Y_{uv}\sqrt{(1-\ell_u)(1-\ell_v)} + (1-r)\,\text{sgn}(b_u b_v)\sqrt{\ell_u \ell_v}}{\sqrt{(1-r\ell_u)(1-r\ell_v)}}.$$

At $r = 1$ the conditional relatives are the summary relatives and Theorem 5 is restored. For non-relatives with $\ell_u, \ell_v \in (0,1)$, $|\pi_{uv}|$ is strictly decreasing in r and vanishes only at $r = 1$. Spurious conditional relatives therefore appear for every $r < 1$ once n is large, in line with known residual-confounding results for unreliable covariates (Westfall & Yarkoni, 2016); fewer than half of the non-relative pairs are declared only if $r > r_{\text{crit}} = 1 - \frac{\tau'(1-\ell)}{\ell(1-\tau')}$, $\tau' = \tanh(z_c/\sqrt{n'-4})$ (equal shares). If the driver reaches nodes through different filters (e.g. travel times k_v), a scalar proxy leaves a nonzero residual even at $r = 1$. The reliability is identified by $\rho_{uZ}\rho_{vZ}/\rho_{uv} = r$ (the classical three-indicator identity; we did not trace its source).

Lean: `prop31_std`, `cor32`, `absPi_strictAntiOn`, `absPi_eq_zero_iff`, `rcrit_iff`, `thm35_bracket`, `pc_lipschitz`, `prop36_ar1_ne_zero`, `prop36_rank_one`, `prop38_ratio`.

Formalisation found four missing nondegeneracy hypotheses in our earlier statements (e.g. $b_u b_v \neq 0$); they are included above (Appendix F). For $\ell = 0.64$ and a typical threshold $\tau' = 0.082$, $r_{\text{crit}} = 0.95$.

The non-asymptotic bounds are loose (factors 3–13 in n, Appendix E); the asymptotic statements matched every simulation to within 0.03 and are used for prediction below.

6. Machine-Checked Proofs

All results of Section 3 and the deterministic and algebraic parts of Sections 4 and 5 are formalised in Lean 4 (de Moura & Ullrich, 2021) with Mathlib (The mathlib Community, 2020) (3,711 lines). The axiom report lists 133 theorems, each depending only on `propext`, `Classical.choice` and `Quot.sound`; the sources contain no `sorry`, `admit` or `axiom`. Relatives, s_ρ, G_T and the stationary windows are defined from scratch following Reisach et al. (2026); "a directed path exists" is read as reflexive-transitive reachability. The 17 newest theorems cover the loop-free edge formula and the accuracy statements. Not formalised: the random-graph limits (Proposition 7), eventual periodicity and the loop-free stationary value, Gaussian tail bounds, Isserlis' theorem, the CLT, the SVAR moving-average representation and genericity (Proposition 8). Appendix F maps every statement to its Lean name and status.

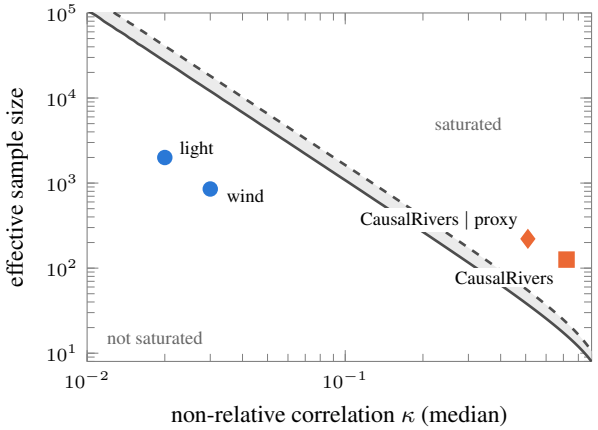


Figure 3. Saturation regime of Proposition 12. Curves: onset $n_{1/2}$ for $z_c = 3.29$ ($d = 5$, solid) and 4.04 ($d = 20$, dashed); above them a non-relative pair is detected with probability $> 1/2$. Points: medians from the real-data studies (CausalRivers unconditioned and given the regional proxy, κ replaced by $|\pi|$; Causal Chambers wind and light tunnel). Values are in Section 7 and Appendix G.

7. Pre-Registered Real-Data Evidence

Protocol. We ran four pre-registered studies (P1–P9, then E1/E2). Each prediction, threshold, estimator, seed and kill rule was committed to version control before any dependence statistic of the data was computed (timeline in Appendix H). The predicted value for an observed node set S is the stationary value (Theorem 5) computed in the full known graph with unobserved nodes latent, averaged over the induced non-self-loop edges ($\tilde{s}$). The estimator is the relatives count of Section 5 at $\alpha = 0.01$ with Bonferroni; orientation accuracy is the mean edge score of $\hat{\rho}$. CIs are cluster bootstraps (2,000 resamples). Lohse & Wahl (2025) measured var- and R^2-sortability on the same families (river variances tend to decrease along the causal order; chamber variances increase); we know of no earlier test of rel-sortability on real data.

CausalRivers: a negative result the theory explains.

Table 2. Pre-registered tests. Pass/fail is by the pre-registered point rule. Brackets: 95% cluster-bootstrap CI. "non-rel. dep.": fraction of graph non-relative pairs declared dependent. "Acc.": orientation accuracy on edges predicted to be strictly sorted.

study (units)	test (threshold)	result (12 h / 24 h)	verdict
CausalRivers, marginal (120 subgraphs)	P1: confluence subgraphs have $\tilde{s} > 1/2$	112 of 114	fails as worded[a]
	P2: Acc. > 0.5, CI excl. 0.5	$0.501\,[0.492, 0.509]$ / $0.498\,[0.489, 0.507]$	**fail**
	P3: Spearman$(\tilde{s}, \text{acc.}) \geq 0.3$	-0.018 / 0.018	**fail**
	(non-rel. dep., nominal test)	0.984 / 0.978	kill rule triggered
CausalRivers, conditioned on regional proxy (120 subgraphs)	P4: non-rel. dep. < 0.5 given proxy	0.741 / 0.735	**fail**
	P5: Acc. > 0.5, CI excl. 0.5	$0.539\,[0.513, 0.565]$ / $0.540\,[0.514, 0.567]$	pass, not interp.[b]
	P6: Spearman ≥ 0.3	-0.113 / -0.121	**fail**
Causal Chambers, wind tunnel (15 runs, primary)	P7: non-rel. dep. < 0.5	$0.039\,[0.031, 0.047]$	pass
	P8: Acc. > 0.5, CI excl. 0.5	$0.656\,[0.637, 0.675]$	pass
	P9: Spearman ≥ 0.3 (300 subgraphs)	$0.267\,[0.163, 0.368]$	**fail**
light tunnel (10 segments, replication)	R7 / R8 / R9 (same thresholds)	0.000 / 1.000 / $0.779\,[0.696, 0.864]$	pass (replication)[c]
E1, light tunnel (10 units, G_{lin})	E1a: non-rel. dep. < 0.5; det. ≥ 0.8	0.000; $0.969\,[0.945, 0.992]$	pass
	E1b: $\Delta \leq -0.10$, Holm $p \leq 0.05$, all 8	$\Delta = -0.502$ to -0.533 (10/10)	pass (8/8)[d]
	E1c: as E1b, detected edges	$\Delta = -0.500$	pass (8/8)
E2, wind regimes (64 units; D=1, lags L=30 vs. 0)	Q7: non-rel. dep. < 0.5	$0.003\,[0.001, 0.005]$	pass
	Q8: Acc. > 0.5, CI excl. 0.5	$0.553\,[0.527, 0.579]$	pass
	Q9: Spearman ≥ 0.3 (1,280 subgr.)	$0.243\,[0.193, 0.294]$	**fail**
	Q10: detection gain, CI lower bound > 0	$+0.069\,[0.060, 0.079]$	pass
	E2 success (Q7–Q10 all)		**not met**

[a]The two exceptions are split-and-rejoin channels with a single source, for which the theorem itself predicts $1/2$. [b]Kill rule (P4 failed at both resolutions): P5 is reported, not interpreted. [c]Pre-registered as not counting towards a positive demonstration (i.i.d. inputs, weak time-series aspect). [d]G_{lin} is the pre-registered primary graph. It omits six real causal edges that, under a symmetry lemma and an assumption of ideal polariser sensors, carry no linear dependence. On the published graph the theorem still beats all eight overall, but not on its strictly ordered edges (Section 7, Appendix I).

River networks are in-forests, so Corollary 6 predicts $\tilde{s} > 1/2$ for subgraphs with confluences (mean 0.82). The estimated relatives sort carried no orientation information (P2, P3 fail; Table 2) because 98% of non-relative station pairs tested dependent. Read through F2 (post hoc for this study): non-relative correlations $\kappa \approx 0.64$–0.74 give $n_{1/2} \approx 20$–27, while the records have $n = 377$–$3{,}652$ steps; the study sat two orders of magnitude inside the saturated regime (Figure 3), where Proposition 12 predicts $\hat{s} = 1/2$. Regional precipitation acts as a latent driver; adding it to the summary graph leaves one source per component, and Corollary 6 then predicts exactly $1/2$. The second study pre-registered the F3 remedy (condition on a leave-out regional mean). It failed (P4: 0.74), and F3 accounts for it: the triad estimate of reliability is $\hat{r} \approx 0.62$, below the $r_{\text{crit}} \approx 0.87$ required at the observed n_{eff} (only 26% of pairs exceed it), and the plug-in prediction of the post-conditioning dependent fraction is 0.743 (0.736) against an observed 0.741 (0.735) at 12 h (24 h). This check is in-sample, because $\hat{r}$ and $\hat{\ell}$ come from the tested correlations. Post hoc, 10 principal components still leave 49% of non-relative pairs dependent, consistent with a driver of rank > 1.

Causal Chambers: designed independent sources. The third study chose data in which Corollary 6's non-trivial case holds by design: in the wind-tunnel random-walk runs (Gamella et al., 2025) five actuators are independent experimenter-generated random walks, so they are separate source SCCs; 25 of 26 edges are predicted to be strictly sorted. Saturation is absent (P7, 0.039) and the relatives count orients confluence edges above chance (P8, 0.656). The graded agreement P9 missed its threshold ($0.267 < 0.3$; CI contains 0.3), so by our own pre-registered definition the study is *not* a positive real-data demonstration. The reason is F1, not a driver: 16 of the 26 ground-truth edges are essentially undetectable at lag 0 at $n \approx 10^3$ (smallest relative correlation ≈ 0.002), which compresses the estimated counts. The light-tunnel replication with i.i.d. inputs passes all three criteria (Table 2). F-theory predictions computed from the first half of each unit (plug-in correlations and autocorrelations, Fisher-z asymptotics of Propositions 11 and 12) and evaluated on the second half are accurate to within 0.02 (Table 3). In particular F2 predicts no saturation: $\hat{\kappa} \approx 0.03$ (wind) and 0.02 (light) give $n_{1/2} \approx 15{,}600$ and 33,800, far above the available $n_{\text{eff}} \approx 430$–$1{,}000$.

A measured driver (ambient pressure, shared by four barometers with no graph edge) is detected exactly where F2 predicts ($\kappa = 0.19$–0.32, $n_{1/2} = 145$–$426 < n_{\text{eff}}$), and conditioning on it removes it (dependent fraction $0.039 \to 0.001$). Caveats: the wind tunnel graph is bipartite, so the prediction coincides with "sensors have more relatives than actuators"

Table 3. Out-of-sample F-theory check on the Causal Chambers: predictions from the first half of each unit, observed rates on the second half. Accuracy is predicted by Monte Carlo of independent pair decisions.

	wind tunnel		light tunnel	
	pred.	obs.	pred.	obs.
non-relatives declared (F2)	0.031	0.028	0.006	0.000
relatives detected (F1)	0.360	0.343	0.679	0.681
orientation accuracy	0.630	0.620	0.999	1.000

(the motivation for E1); all units of a chamber share one device.

E1: does the theorem beat naive predictors? Preregistered (Appendix H) to answer the bipartite caveat. The test unit is an edge instance: the theorem's value $s_\rho \in \{0, \frac{1}{2}, 1\}$ against the observed value from the estimated counts, with absolute loss, on instances where the theorem and a competitor disagree. The eight competitors are the sensor indicator; ancestor count, in-degree and depth (each on the full and the induced graph); and the finite-T limit "every edge is 1". A competitor is beaten if $\Delta \le -0.10$ with an exact sign-flip test over units and Holm correction. The primary data are the light-tunnel series `citris_1` (10 units), the only real time series we found with several ground-truth layers among measured variables. **It passes 8/8**: the power prerequisites hold (non-relatives declared 0.000, relatives detected 0.969), $\Delta = -0.502$ to -0.533 in 10/10 units, Holm $p = 0.0078$ (the smallest attainable). The theorem's value equals the observed one on 99.2% of 1,662 instances, the competitors' on 11–20%. The same test on 45 i.i.d. structural-model units (8,734 instances) also passes 8/8; the chain gives $1/2$ everywhere and the collider 1 into the collider, as Corollary 6 predicts. Four statements bound this result (details and numbers in Appendix I). *(i) The primary prediction uses a refined graph.* It removes six real causal edges because, by a symmetry lemma that assumes ideal polariser sensors, they carry no linear dependence; the claim is only that the theory predicts the *linearly dependent* relatives. On the published graph the theorem still beats all eight overall ($\Delta = -0.32$ to -0.42), but on the 442 edges where it predicts strict order the sensor indicator is significantly better ($\Delta = +0.453$, reverse Holm $p = 0.002$), and the theorem's exact-match rate (71.4%) is below that of a constant-$1/2$ rule (81.5%, post hoc). *(ii) Most of the separation comes from ties*: 1,365 of the 1,662 predictions are $1/2$, which no naive rule can express; the 297 strict predictions are also won against all eight. A constant-$1/2$ rule matches 81.5% of `citris_1` against 99.2% (post hoc). *(iii) One device in a favourable regime* (detection 0.97). *(iv) The wind tunnel does not replicate the strength*: the theorem passes the same test on 64 held-out regimes ($\Delta = -0.225$), but detection is only 0.33–0.59, and on the primary configuration a constant-$1/2$ rule has lower absolute error (0.311 vs. 0.363).

E2: lagged relatives. E2 tests whether a relatives test using cross-correlations at lags $|h| \le L$ (Bonferroni over lags; Appendix J) recovers the weak lag-0 edges behind P9, on 64 held-out regimes (primary: first differences, $L = 30$ against $L = 0$). Success required Q7–Q10. **It does not succeed**: Q9 fails (Table 2). False positives stay rare (Q7 0.003), accuracy is 0.553 (Q8), detection rises from 0.328 to 0.398 (Q10), but the graded agreement is $0.243 < 0.3$ (Q9). The secondary configuration (block means, $L = 3$) fails Q9 and Q10 (-0.007): the Bonferroni cost exceeds the gain. The lag-0 estimator rerun on these regimes again fails the graded criterion (0.202 at D=10, 0.120 at D=1), so P9's verdict (0.267) stands; a redraw of the subgraph sample moved that statistic to 0.338 (development, not a verdict), which we report as variability of about 0.07, not as a pass. Every microphone edge stays undetected, so the F1 shortfall is not mainly a lag problem.

8. Discussion and Limitations

For benchmark design. The finite time-unrolled generator is rel-sortable above $1/2$ for every summary graph that is not a union of strongly connected components (Corollary 3), and part of the excess comes only from the start at $t = 1$. Benchmarks should use strongly connected summary graphs or a burn-in of at least $p - 1$ slices, after which the value is the graph property of Theorem 5. Random summary graphs are sortable only near criticality (Figure 2).

For real data. Population rel-sortability above $1/2$ does not imply that a relatives count can be estimated. Two conditions are needed: no latent common driver with share ℓ such that $n \gtrsim z_c^2/\ell^2$ (or a near-perfect, rank-one proxy), and relatives detectable at lag 0 at the available n_{eff}. CausalRivers violates the first, the wind tunnel partly the second. Diagnostic: compare n_{eff} with $3 + (z_c/\operatorname{atanh}\hat{\kappa})^2$, where $\hat{\kappa}$ is the typical correlation of pairs believed unrelated; if n_{eff} is far above it, the count is saturated whatever the graph.

Open questions. Proofs of the random-graph limits and of the windowed loop-free stationary value; contemporaneous effects; nonlinear mechanisms (faithfulness can fail, as the Malus-law pairs of the light tunnel show).

Limitations. The closed forms assume self-loops at every node (without them a single edge gives $1/2$ instead of $2/3$; Section 3.1). The ER limits are a sketch with numerical support. Section 4 covers linear SVARs only. The finite-sample bounds are loose, and the asymptotic statements rest on an assumed joint CLT. Proposition 8 and the probabilistic parts of F1–F3 are proved on paper only. The positive real-data evidence comes from one device per chamber; E1 holds for linearly dependent relatives, assumes ideal polariser sensors, and is carried mainly by predicted ties. The graded criteria P2, P3, P4, P6, P9 and Q9 failed.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

Artefact Statement

Lean code, verification scripts, analysis scripts, preregistration documents and run logs are provided as supplementary material (anonymised). Raw data are public (CausalRivers; Causal Chambers, CC BY 4.0) and are not redistributed.

References

Afyouni, S., Smith, S. M., and Nichols, T. E. Effective degrees of freedom of the Pearson's correlation coefficient under autocorrelation. *NeuroImage*, 199:609–625, 2019.

Bartlett, M. S. Some aspects of the time-correlation problem in regard to tests of significance. *Journal of the Royal Statistical Society*, 98(3):536–543, 1935.

Clifford, P., Richardson, S., and Hémon, D. Assessing the significance of the correlation between two spatial processes. *Biometrics*, 45(1):123–134, 1989.

de Moura, L. and Ullrich, S. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, Lecture Notes in Computer Science, pp. 625–635, 2021.

Dorogovtsev, S. N., Mendes, J. F. F., and Samukhin, A. N. Giant strongly connected component of directed networks. *Physical Review E*, 64(2):025101, 2001.

Gamella, J. L., Peters, J., and Bühlmann, P. Causal chambers as a real-world physical testbed for AI methodology. *Nature Machine Intelligence*, 7(1):107–118, 2025.

Ginovyan, M. S. and Taqqu, M. S. A survey on limit theorems for Toeplitz type quadratic functionals of stationary processes and applications, 2021.

Gong, M., Zhang, K., Schölkopf, B., Tao, D., and Geiger, P. Discovering temporal causal relations from subsampled data. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pp. 1898–1906, 2015.

Gong, M., Zhang, K., Schölkopf, B., Glymour, C., and Tao, D. Causal discovery from temporally aggregated time series. In *Proceedings of the 33rd Conference on Uncertainty in Artificial Intelligence*, 2017.

Herman, R. J., Wahl, J., Ninad, U., and Runge, J. Unitless unrestricted Markov-consistent SCM generation: Better benchmark datasets for causal discovery. In *Proceedings of the Fourth Conference on Causal Learning and Reasoning*, volume 275 of *Proceedings of Machine Learning Research*, pp. 1506–1531, 2025.

Hsu, D., Kakade, S. M., and Zhang, T. A tail inequality for quadratic forms of subgaussian random vectors. *Electronic Communications in Probability*, 17, 2012. arXiv:1110.2842.

Laurent, B. and Massart, P. Adaptive estimation of a quadratic functional by model selection. *The Annals of Statistics*, 28(5):1302–1338, 2000.

Lohse, C. and Wahl, J. Sortability of time series data. *Transactions on Machine Learning Research*, 2025. arXiv:2407.13313.

Ormaniec, W., Sussex, S., Lorch, L., Schölkopf, B., and Krause, A. Standardizing structural causal models. In *International Conference on Learning Representations*, 2025.

Reisach, A. G., Seiler, C., and Weichwald, S. Beware of the simulated DAG! Causal discovery benchmarks may be easy to game. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

Reisach, A. G., Tami, M., Seiler, C., Chambaz, A., and Weichwald, S. A scale-invariant sorting criterion to find a causal order in additive noise models. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Reisach, A. G., Chambaz, A., Blanchard, G., and Weichwald, S. A topological sorting criterion for random causal directed acyclic graphs. arXiv:2605.06288, 2026.

Roy, R. Asymptotic covariance structure of serial correlations in multivariate time series. *Biometrika*, 76(4): 824–827, 1989.

Runge, J., Nowack, P., Kretschmer, M., Flaxman, S., and Sejdinovic, D. Detecting and quantifying causal associations in large nonlinear time series datasets. *Science Advances*, 5(11):eaau4996, 2019.

Stein, G., Shadaydeh, M., Blunk, J., Penzel, N., and Denzler, J. CausalRivers – scaling up benchmarking of causal discovery for real-world time-series. In *International Conference on Learning Representations*, 2025.

Tan, J. and Syrgkanis, V. CausalSmith: A formally grounded, self-improving agentic framework for automated research in causal inference, 2026.

The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pp. 367–381, 2020.

Westfall, J. and Yarkoni, T. Statistically controlling for confounding constructs is harder than you think. *PLOS ONE*, 11(3):e0152719, 2016.

Zheng, X., Aragam, B., Ravikumar, P., and Xing, E. P. DAGs with NO TEARS: Continuous optimization for structure learning. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Appendix overview

Appendix A gives full proofs of Section 3, Appendix B the exhaustive enumeration check, Appendix C the proofs of Section 4, Appendix D the proofs of Section 5, Appendix E the simulation checks of F1–F3, Appendix F the map from statements to Lean names, Appendix G details of the real-data studies, Appendix H the pre-registration timeline and deviations, Appendices I and J the experiments E1 and E2 in full, and Appendices K to M the loop-free graphs, the random summary graphs and the accuracy interpretation.

Status labels. *Machine-checked*: a Lean theorem with no `sorry` and only the standard axioms. *Proved*: a complete paper proof given here. *Asymptotic*: the argument assumes a joint central limit theorem for the sample variances and cross-covariance of a bivariate stationary Gaussian process; scalar versions are surveyed in Ginovyan & Taqqu (2021), and we do not prove the joint statement (a Lindeberg argument for the weighted chi-square sums is possible but is not written out here). *Sketched*: a derivation with one analytic step left out, stated as such. *Enumerated*: checked by exact computation on many instances, not proved.

A. Proofs for Section 3

Throughout, $G_{\mathrm{S}} = (V, E_{\mathrm{S}})$ has a self-loop at every node, $p = |V| \geq 2$, and $d(u, v)$ is as in Section 2. We use two facts of Reisach et al. (2026), proved there for DAGs: (Thm. 1) if $x \in \mathrm{anc}(y)$ then $\mathrm{rel}(x) \subseteq \mathrm{rel}(y)$; (Lemma 1, eq. (10)) if $x \in \mathrm{anc}(y)$ then $\mathrm{rel}(x) \subsetneq \mathrm{rel}(y)$ iff $\mathrm{roots}(y) \setminus \mathrm{roots}(x) \neq \emptyset$, where $\mathrm{roots}(y) = \{x \in \mathrm{anc}(y) : \mathrm{anc}(x) = \{x\}\}$. Both are also machine-checked independently (`relSet_subset_of_edge`, and the direct characterisation `isRel_unrolled_iff`).

Lemma 14 (ancestors in G_T)**.** *$u(s) \in \mathrm{anc}(v(t))$ iff $s \leq t$ and $d(u, v) \leq t - s$.*

Lean: `unrolled_reach_iff`, `unrolled_reach_iff_dist`.

Proof. Every edge of G_T increases time by one, so a directed path from $u(s)$ to $v(t)$ has $t - s$ edges, and its projection to V is a walk of length $t - s$ from u to v in G_{S} (self-loops allowed as steps). Such a walk exists iff $d(u, v) \leq t - s$: a walk contains a path, so its length is at least $d(u, v)$; conversely, pad a shortest path with $t - s - d(u, v)$ self-loop steps at u. Every such walk lifts to a directed path in G_T. □

Lemma 15 (roots)**.** $\mathrm{roots}(v(t)) = \{u(1) : d(u, v) \leq t - 1\}$, *and* $\mathrm{desc}(u(1)) = \{w(s) : d(u, w) \leq s - 1\}$.

Lean: `roots_unrolled_iff`, `isRoot_unrolled_iff`, `desc_root_unrolled_iff`.

Proof. A node $u(s)$ with $s \geq 2$ has the parent $u(s - 1)$ through the self-loop, so it is not a root; $u(1)$ has no parent. Apply Lemma 14 with $s = 1$, in both directions. □

Proof of Theorem 1. $i(t)$ is a parent of $j(t{+}1)$, so $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$ and $s_\rho(e) \in \{1/2, 1\}$, with value 1 iff the inclusion is strict, iff $\mathrm{roots}(j(t{+}1)) \setminus \mathrm{roots}(i(t)) \neq \emptyset$. By Lemma 15 this set is $\{u(1) : d(u, j) \leq t,\ d(u, i) > t - 1\}$, and since d is integer-valued or ∞, $d(u, i) > t - 1$ iff $d(u, i) \geq t$. □

Proof of Proposition 4 (and the general count). Since every ancestor of y descends from a root of y that is itself an ancestor, $\mathrm{rel}(y) = \mathrm{desc}(\mathrm{roots}(y))$. By Lemma 15, $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d(u, v) \leq t - 1$ and $d(u, w) \leq s - 1$, i.e. iff $s \geq 1 + m_t(v, w)$ with $m_t(v, w) = \min\{d(u, w) : d(u, v) \leq t - 1\}$. Counting $s \in \{1, \ldots, T\}$ gives $\rho(v(t)) = \sum_w \max(T - m_t(v, w), 0)$ (`rho_unrolled_eq`). For $t \geq p$ every finite $d(u, v)$ is at most $p - 1 \leq t - 1$, so the admissible u are $\mathrm{anc}_S(v)$ and $m_t(v, w) = d(\mathrm{anc}_S(v), w)$, finite iff $w \in \mathrm{rel}_S(v)$ and then at most $p - 1 < t \leq T$. (The Lean statement `rho_unrolled_large_t` needs only $t \geq p$; an earlier hypothesis $T \geq p$ is redundant.) □

Proof of Theorem 2. Crossing edge $(i, j) \in E_\times$: j does not reach i, so $d(j, i) = \infty$, and $u = j$ gives $d(j, j) = 0 \leq t \leq \infty$ for every t: all $T - 1$ copies score 1. Edge in $E_\circ$: i and j reach each other (or $i = j$), so for every u, $d(u, i) < \infty$ iff $d(u, j) < \infty$. If the condition of Theorem 1 held with $d(u, i) = \infty$ then $d(u, j) = \infty > t$, a contradiction; hence it forces $t \leq d(u, i) \leq p - 1$, and at most $N_T(i, j) \leq \min(p - 1, T - 1)$ copies score 1. Summing over the $(T - 1)|E_{\mathrm{S}}|$ edges,

$$s_\rho(G_T) = \tfrac{1}{2} + \frac{1}{2(T-1)|E_{\mathrm{S}}|}\Big[(T-1)|E_\times| + \sum_{(i,j)\in E_\circ} N_T(i, j)\Big],$$

which gives both bounds; let $T \to \infty$. □

Proof of Corollary 3. $E_\times = \emptyset$ iff no edge joins two SCCs. A weakly connected component made of two or more SCCs is connected by undirected edges, one of which joins two different SCCs; conversely a strongly connected component has no crossing edge. The uniform bound is Theorem 2 with $|E_\times| \geq 1$. For the DAG case, every SCC is a singleton, $E_\times$ consists of the m non-loop edges, and $|E_S| = m + p$. □

Proof of Theorem 5. In G_{B+T}, by Lemma 15 shifted by B, $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d(u,v) \leq t-1$ and $d(u,w) \leq s-1$. For window nodes $t-1, s-1 \geq B \geq p-1$, and every finite distance is at most $p-1$, so the condition is $u \in \mathrm{anc}_S(v) \cap \mathrm{anc}_S(w)$, independent of s,t. Hence $\mathrm{rel}_W(v(t)) = \{w(s) \in W : w \in \mathrm{rel}_S(v)\}$, of size $T|\mathrm{rel}_S(v)|$. On $V \times \mathbb{Z}$, Lemma 14 holds verbatim and the common ancestor can be taken arbitrarily early, giving the same set. Each copy of (i,j) therefore compares $|\mathrm{rel}_S(i)|$ with $|\mathrm{rel}_S(j)|$; as $i \in \mathrm{anc}_S(j)$, $\mathrm{rel}_S(i) \subseteq \mathrm{rel}_S(j)$, so the score is 1 iff $\mathrm{rel}_S(i) \neq \mathrm{rel}_S(j)$, and $1/2$ for self-loops. Source SCCs: $\mathrm{rel}_S(j) \setminus \mathrm{rel}_S(i) \neq \emptyset$ iff $\mathrm{anc}_S(j) \not\subseteq \mathrm{rel}_S(i)$ (if $z \in \mathrm{rel}_S(i)$ then $\mathrm{desc}_S(z) \subseteq \mathrm{rel}_S(i)$). Every node of $\mathrm{anc}_S(j)$ is reached from some $\sigma \in \mathrm{Src}(j)$, and $\mathrm{desc}_S(\sigma) \subseteq \mathrm{rel}_S(i)$ whenever $\sigma \subseteq \mathrm{rel}_S(i)$. A source SCC meets $\mathrm{rel}_S(i)$ iff it reaches i (its only ancestors are its own nodes), and then it is contained in $\mathrm{rel}_S(i)$. So the condition is $\mathrm{Src}(j) \not\subseteq \mathrm{Src}(i)$, which, as $\mathrm{Src}(i) \subseteq \mathrm{Src}(j)$, is $\mathrm{Src}(i) \neq \mathrm{Src}(j)$. □

Proof of Corollary 6. If each weak component C has a unique source SCC σ_C, then $\mathrm{Src}(v) = \{\sigma_C\}$ on C and $E_\uparrow = \emptyset$. Conversely, if $E_\uparrow = \emptyset$, Src is constant along edges and hence on each weak component; a node v of a source SCC σ has $\mathrm{Src}(v) = \{\sigma\}$, so there is only one. Out-forests (at most one parent besides itself; acyclicity is not needed, `stat_half_of_outForest`) and strongly connected graphs (`stat_half_of_strong`) have one source SCC per component. In an in-forest (acyclic apart from self-loops, out-degree at most one), if $u \in \mathrm{anc}(v)$ then $\mathrm{desc}(u)$ is the unique downstream path through v, so $\mathrm{rel}(v) = \mathrm{anc}(v) \cup \mathrm{desc}(v)$ (`inForest_isRel_iff`); the source SCCs are the headwaters, and $\mathrm{Src}(j) = \mathrm{Src}(i) \cup \bigcup_k \mathrm{Src}(k)$ over the other parents k of j, with disjoint headwater sets on different branches, so $\mathrm{Src}(j) \neq \mathrm{Src}(i)$ iff j is a confluence (`inForest_up_iff`). Out-degree at most one alone is not enough: for $a \rightleftarrows b$ with $c \to a$, a has two parents but the only source SCC is $\{c\}$. □

Remark 16 (self-loops). Without self-loops the lemmas fail: for the single edge $0 \to 1$ without loops, $s_\rho(G_T) = 1/2$ for every $T \geq 2$ (`edgeNoLoops_sRho`); with loops the limit is $2/3$ (`edgeLoops_limit`).

B. Exhaustive Enumeration Check

Before formalisation, every statement of Section 3 was checked by brute force (bitset relatives with the definitions of Reisach et al., 2026) on all $4 + 64 + 4096$ summary graphs with $p = 2, 3, 4$ (all $T \leq 9$) and on 10,000 random graphs ($p = 2$–7; Erdős–Rényi, DAG, in-forest, out-forest, chain with back edge and block-SCC families; three values of $T \leq 14$ each, fixed seed). In total 46,724 (graph, T) cases and 1,895,609 edges were checked with zero failures of Theorem 1, Proposition 4, the bounds of Theorem 2, Theorem 5 (with $B = p-1$ and $B = 2p+1$) and the source-SCC characterisation. The conjecture we had pre-registered ("an edge is strictly sorted in the limit iff $|\mathrm{rel}_S(j)| > |\mathrm{rel}_S(i)|$") disagreed with the exact G_T limit in 9,603 random graphs and never with the stationary value; the conjecture is Theorem 5, not the limit of G_T.

C. Proofs for Section 4

Assumptions: $X(t) = \sum_{\tau=1}^{L} A_\tau X(t-\tau) + \varepsilon(t)$, $t \in \mathbb{Z}$, i.i.d. innovations with diagonal positive-definite covariance Σ, stable (all roots of $\det(I - \sum_\tau A_\tau z^\tau)$ outside the unit disc), stationary solution; the summary graph has $i \to j$ iff $(A_\tau)_{ji} \neq 0$ for some τ.

Proof of Proposition 8. By the MA(∞) representation $X(t) = \sum_k \Psi_k \varepsilon(t-k)$, and because Ψ_k is supported on the reachability relation, $X_i(t)$ is a measurable function of $\{\varepsilon_k(s) : k \in \mathrm{anc}_S(i)\}$. Disjoint ancestor sets mean independent innovation sets, so the processes X_i and X_j are independent, and so are any per-node transforms. For the generic part, each covariance entry is real-analytic in (A, Σ) on the connected open set of stable parameters with the given support, hence either identically zero or nonzero almost everywhere. With all coefficients positive, including the self-loop at a common ancestor k, all Ψ entries are nonnegative and $\Psi_{ik}(m), \Psi_{jk}(m) > 0$ for all large m (self-loops pad the walks), so $\Gamma(0)_{ij} > 0$; the same holds for $\Gamma(h)$, $h \geq 0$, and therefore for the Fejér sums and S_{LR}. "Generic" does not mean "large":

with little autocorrelation, $\Gamma(0)_{ij}$ along a lagged edge can be tiny (down to 10^{-14} in our exact computations; exactly 0 without autocorrelation). □

Proof of Theorem 9. (a) $D^2\mathrm{Cov}(\bar{Y}) = \sum_{u,w<D}\Gamma(u-w) = D\Gamma(0) + \sum_{h=1}^{D-1}(D-h)(\Gamma(h)+\Gamma(-h))$ (the counting identity is `fejer_identity`). The stable VAR has $\|\Gamma(h)\| \le Cr^{|h|}$, $r<1$, so $\sum_h |h|\|\Gamma(h)\| < \infty$ and $S_{LR} = \sum_h \Gamma(h) = \Psi(1)\Sigma\Psi(1)^\top$ with $\Psi(1) = (I-A(1))^{-1}$. Then $S_{LR} - D\,\mathrm{Cov}(\bar{Y}) = \sum_{|h|\ge D}\Gamma(h) + \sum_{|h|<D}\frac{|h|}{D}\Gamma(h)$, and $1 \le |h|/D$ in the first sum. (b) Write $I-A(1) = (I-\Lambda) - N$ with Λ the diagonal and N the off-diagonal part; then $I-A(1) = (I-\Lambda)(I-\tilde{B})$ with $\tilde{B} = (I-\Lambda)^{-1}N$ strictly triangular in a topological order (`longrun_factor`, `longrun_inv`). Under acyclicity the AR polynomial factorises into univariate $\phi_j(z) = 1 - \sum_\tau (A_\tau)_{jj} z^\tau$; stability, $\phi_j(0) = 1$ and the intermediate value theorem give $\phi_j(1) = 1-\lambda_j > 0$ (`selfloop_sum_lt_one`). (c) Stationarity. (d) For a linear SEM $X = BX + e$ with noise variances ω, $R_i^2 = 1 - 1/(\Sigma_{ii}(\Sigma^{-1})_{ii})$ and $(\Sigma^{-1})_{ii} = 1/\omega_i + \sum_{k\in\mathrm{ch}(i)} B_{ki}^2/\omega_k$. In an out-star with root 0 and leaves k, $1-R_0^2 = 1/(1+\sum_k r_k)$ and $1-R_k^2 = 1/(1+r_k)$ with $r_k = \omega_0 B_{k0}^2/\omega_k > 0$, so with ≥ 2 leaves the root has strictly larger R^2 than every leaf; the in-star is symmetric. Strict inequalities in the limit SEM of (b), plus (a) and continuity of $S \mapsto R^2$ at positive-definite S, give the claim for all large D. Under subsampling every contemporaneous statistic is a function of $\Gamma(0)$. □

The R^2 statement for stars strengthens the three-node analysis in Herman et al. (2025, App. D), whose qualitative message (that R^2 reflects connectivity, with an asymmetry between colliders and confounders) we credit. Numerical observations (exact population moments, not proved): mean block-mean R^2-sortability over 150 in-trees moves from 0.853 ($D=1$) to 0.834 ($D=4$–8) and 0.880 ($D=256$), so it is not monotone in D; on out-stars subsampled data are R^2-sorted at 0.842 on average while block means give 0.546 ($D=4$), 0.101 ($D=16$) and 0 in the limit.

D. Proofs for Section 5

Notation. $\hat{C}_{ab} = \frac{1}{n}\sum_t (X_a(t)-\bar{X}_a)(X_b(t)-\bar{X}_b)$, $\hat{r}_{ab} = \hat{C}_{ab}/\sqrt{\hat{C}_{aa}\hat{C}_{bb}}$. Constants: B_{ab} as in the main text; $D_\gamma = \max_a 2\sum_h \gamma_{aa}(h)^2$; $\kappa_1 = \max_a \sum_h |\gamma_{aa}(h)|$; $\kappa_2 = \max_{a\ne b}(\sum_h |\gamma_{aa}(h)| + \sum_h |\gamma_{ab}(h)|)$. For i.i.d. sampling, $B_{ab} = 1+\rho_{ab}^2$, $D_\gamma = 2$, $\kappa_1 = 1$.

Lemma 17 (Gaussian chaos tail)**.** *Let $g \sim N(0, I_N)$, A symmetric, $S = g^\top Ag - \mathrm{tr}A$. For $x>0$, $P(S \ge 2\|A\|_F\sqrt{x} + 2\|A\|_{op}x) \le e^{-x}$, and the same for $-S$.*

Lean: algebraic steps `phi_le`, `chernoff_logmgf_le`, `chernoff_choice`, `sqrt_two_v_x`; the probability statement is not machine-checked.

Proof. Diagonalise; S has the law of $\sum_i \lambda_i(g_i^2-1)$. For $2s\lambda < 1$, $\log Ee^{s\lambda(g^2-1)} = \varphi(s\lambda)$ with $\varphi(u) = -\frac{1}{2}\log(1-2u) - u = \sum_{k\ge 2}(2u)^k/(2k) \le u^2/(1-2|u|)$ for $|u| < 1/2$. With $c = 2\|A\|_{op}$, $v = 2\|A\|_F^2$ and $0 \le s < 1/c$, $\log Ee^{sS} \le vs^2/(2(1-cs))$. Markov's inequality with $t = \sqrt{2vx} + cx$, $q = \sqrt{2x/v}$ and $s = q/(1+cq)$ gives exponent exactly x. This is the Gaussian case of Hsu et al. (2012, Prop. 1), who derive it from Laurent & Massart (2000); the proof here is self-contained. □

Lemma 18 (exact variance)**.** *With $\tilde{\gamma}_{ab} = \frac{1}{n}\sum_t X_a(t)X_b(t) = g^\top Ag$ for $A = C^{1/2}MC^{1/2}$, $M = \frac{1}{2n}\left(\begin{smallmatrix}0 & I\\ I & 0\end{smallmatrix}\right)$,*

$$2n\|A\|_F^2 = \sum_{|h|<n}\Big(1-\tfrac{|h|}{n}\Big)\big[\gamma_{aa}(h)\gamma_{bb}(h) + \gamma_{ab}(h)\gamma_{ab}(-h)\big] \le B_{ab},$$

$\|A\|_{op} \le \kappa_2/(2n)$; on the diagonal $2n\|A\|_F^2 \le D_\gamma$ and $\|A\|_{op} \le \kappa_1/n$; and $\bar{X}_a \sim N(0, \bar{\sigma}_a^2)$ with $\bar{\sigma}_a^2 \le \kappa_1/n$.

Lean: trace and Toeplitz algebra `quad_M`, `trace_MC`, `lemma12_trace`, `lemma12_fejer`, `lemma12_le_B`, `lemma12_diag_le_D`; the step $\|C^{1/2}MC^{1/2}\|_F^2 = \mathrm{tr}(MCMC)$ and the operator-norm bounds are not machine-checked.

Proof. $\|A\|_F^2 = \mathrm{tr}(MCMC)$ and, with Toeplitz blocks $(C_{ab})_{st} = \gamma_{ab}(s-t)$, $4n^2\mathrm{tr}(MCMC) = 2\sum_{s,t}[\gamma_{aa}(s-t)\gamma_{bb}(s-t) + \gamma_{ab}(s-t)\gamma_{ab}(t-s)]$ (using that γ_{aa}, γ_{bb} are even); count pairs with $s-t=h$. $\|A\|_{op} \le \|C\|_{op}\|M\|_{op}$ and $\|C\|_{op}$ is at most the largest absolute row sum. □

Lemma 19 (deviation event). *Let* $\varepsilon_{ab} = \sqrt{2B_{ab}x/n} + \kappa_2 x/n + 2\kappa_1 y/n$ *and* $\varepsilon_d = \sqrt{2D_\gamma y/n} + 4\kappa_1 y/n$. *With probability at least* $1 - d(d-1)e^{-x} - 4de^{-y}$, $|\hat{C}_{ab} - \rho_{ab}| \le \varepsilon_{ab}$ *for all* $a \neq b$ *and* $|\hat{C}_{aa} - 1| \le \varepsilon_d$ *for all* a.

Lean: deterministic steps `centred_cov_eq`, `centred_dev_offdiag`, `centred_dev_diag`, `lemma13_eps`, `lemma13_count`.

Proof. $\hat{C}_{ab} = \tilde{\gamma}_{ab} - \bar{X}_a \bar{X}_b$. Apply Lemmas 17 and 18 to each off-diagonal and diagonal $\tilde{\gamma}$, bound $|\bar{X}_a| \le \sqrt{2\kappa_1 y/n}$ by the Gaussian Chernoff bound, and take a union bound. □

Lemma 20 (correlation perturbation). *If* $a', b' \in [1-\varepsilon_d, 1+\varepsilon_d]$, $\varepsilon_d < 1$ *and* $|c' - \rho| \le \varepsilon$, *then* $|c'/\sqrt{a'b'} - \rho| \le (\varepsilon + |\rho|\varepsilon_d)/(1-\varepsilon_d)$.

Lean: `corr_perturb` *(the hypothesis* $|\rho| \le 1$ *of our earlier statement is not needed).*

Proof. $|c'/\sqrt{a'b'} - \rho| \le |c'-\rho|/\sqrt{a'b'} + |\rho||1/\sqrt{a'b'} - 1|$, with $1/\sqrt{a'b'} \le 1/(1-\varepsilon_d)$ and $|1/\sqrt{a'b'} - 1| \le \varepsilon_d/(1-\varepsilon_d)$. □

Proof of Theorem 10. Set $\delta_{ab} = (\varepsilon_{ab} + |\rho_{ab}|\varepsilon_d)/(1-\varepsilon_d)$. On the event of Lemma 19, Lemma 20 gives $|\hat{r}_{ab} - \rho_{ab}| \le \delta_{ab}$, so non-relatives satisfy $|\hat{r}| \le \tau$ and relatives $|\hat{r}| > \tau$ (`decide_eq_target`, `thm15_det`). Equal declared sets give equal counts and hence $\hat{s} = s_\rho^{\mathrm{pop}}$ (`sOf_cntDecl_congr`). For a VAR observed as a long stationary series, s_ρ^{pop} is the stationary value of Theorem 5 on non-self-loop edges (Proposition 8). Explicit form: with $y = x + \log(4/(d-1))_+$ (`explicit_y`) and a common threshold $\tau < \rho_{\min}$, $P(\hat{R} \neq R) \le 2d(d-1)\exp\big(-n\min(\tau, \rho_{\min} - \tau)^2/(2(\sqrt{B} + \rho_{\max}\sqrt{D_\gamma})^2)(1+o(1))\big)$. □

Argument for Proposition 11 (asymptotic). By Isserlis' theorem, $n\,\mathrm{Cov}(\tilde{\gamma}_{ab}, \tilde{\gamma}_{cd}) \to V(ab,cd) = \sum_h [\gamma_{ac}(h)\gamma_{bd}(h) + \gamma_{ad}(h)\gamma_{bc}(h)]$ (as in Lemma 18); centring is $O(1/n)$. Assuming the CLT for quadratic forms, the delta method with gradient $(-\rho/2, -\rho/2, 1)$ gives $\sqrt{n}(\hat{r} - \rho) \Rightarrow N(0, W)$, $W = V(ab,ab) - \rho[V(ab,aa) + V(ab,bb)] + \frac{\rho^2}{4}[V(aa,aa) + 2V(aa,bb) + V(bb,bb)]$, which equals B_{ab} for an independent pair. Then $\sqrt{n}\,\operatorname{atanh}\hat{r} \Rightarrow N(0, B_{ab})$ under the null and $P(|\operatorname{atanh}\hat{r}| > z_c/\sqrt{n-3}) \to 2\bar{\Phi}(z_c/\sqrt{B_{ab}})$. □

Proof of Proposition 12. $\mathrm{Cov}(X_u, X_v) = \gamma^Y_{uv}(0) + b_u b_v$ and $\gamma^Y_{uv} = 0$ for non-relatives (Proposition 8); divide by the standard deviations (`prop21_corr_std`). For relatives the sum is nonzero except on a Lebesgue-null set. All counts equal d, so every edge scores $1/2$ (`sOf_saturated`). The rate: apply Theorem 10 with target "all pairs" and $x = y = cn$, c small enough that $\delta \le \kappa_{\min}/3$ and $\tau_n \le \kappa_{\min}/3$ eventually; $|\hat{s} - 1/2| \le 1/2$ always (`abs_sOf_sub_half_le`). The onset: by the asymptotic normality of $\operatorname{atanh}\hat{r}$, the declaration probability crosses $1/2$ where $\tanh(z_c/\sqrt{n-3}) = |\kappa|$, i.e. at $n_{1/2}$ (`ncrit_eq_iff`). □

Proof of Proposition 13. $\mathrm{Cov}(X_u, Z) = b_u\sqrt{r}$ and $\mathrm{Var}Z = 1$; the partial covariance is $\gamma^Y_{uv} + (1-r)b_u b_v$ and the partial variance $\mathrm{Var}Y_u + (1-r)b_u^2$; divide and standardise (`prop31`, `prop31_std`; positive $\mathrm{Var}Y_u$ is required). At $r = 1$, $\pi = \rho^Y$ (`cor32`). For non-relatives $|\pi_{uv}| = \sqrt{\ell_u \ell_v}\sqrt{f_u f_v}$ with $f_w(r) = (1-r)/(1-r\ell_w)$ and $f'_w(r) = (\ell_w - 1)/(1-r\ell_w)^2 < 0$ (`absPi_strictAntiOn`, `absPi_eq_zero_iff`; both need $b_u b_v \neq 0$). Bounds $(1-r)\sqrt{\ell_u\ell_v} \le |\pi| \le (1-r)\sqrt{\ell_u\ell_v}/(1-\ell_{\max})$ (`absPi_lower`, `absPi_upper`). r_{crit} solves $(1-r)\ell/(1-r\ell) = \tau'$ (`rcrit_iff`, `rcrit_example`). The finite-sample bracket uses the Lipschitz bound $|g(a',b',c') - g(a,b,c)| \le L(q)\max(|a'-a|, |b'-b|, |c'-c|)$ for $g(a,b,c) = (a-bc)/\sqrt{(1-b^2)(1-c^2)}$, $|b|, |c| \le q < 1$, $L(q) = 1/(1-q^2) + 4q/(1-q^2)^2$ (`pc_lipschitz`, `thm35_bracket`), combined with Lemma 19 for (X, Z). Multi-filter driver: for $D_v(t) = b_v C(t - k_v)$ with C an AR(1) with coefficient ϕ and $Z = C(t)$,

$$\pi_{uv} = \frac{b_u b_v(\phi^{|k_u - k_v|} - \phi^{k_u + k_v})}{\sqrt{(s_u^2 - b_u^2\phi^{2k_u})(s_v^2 - b_v^2\phi^{2k_v})}},$$

nonzero for $k_u, k_v > 0$, $\phi \in (0,1)$, $b_u b_v \neq 0$ (`prop36_ar1`, `prop36_ar1_ne_zero`); in general $\pi_{uv} \propto \mathrm{Cov}(D_u - P_Z D_u, D_v - P_Z D_v)$ vanishes for all non-relative pairs iff the residual driver components are uncorrelated, which holds for a rank-one driver with $Z = F$ (`prop36_pcov`, `prop36_zero_iff`, `prop36_rank_one`). A proxy that contains summary relatives of u, v adds the terms $-c_Z(b_u q_v + b_v q_u) - q_u q_v$ to the numerator (`prop37_pcov`, `prop37_bound`). Triad identity: $\rho_{uZ}\rho_{vZ}/\rho_{uv} = r$ and, for $r > 0$, $\ell_u = \rho^2_{uZ}/r$ (`prop38_ratio`, `prop38_loading`). □

E. Simulation Checks of F1–F3

All with fixed seeds; full tables are in the supplementary material. **F1** (5 processes: in-forests with $d = 5$ and $d = 10$, self-loop persistence $\lambda \in \{0.5, 0.9, 0.97\}$, and a near-white-noise case; $n = 200$–$20{,}000$; 1,000 replications per cell; nominal, oracle-effective and fixed thresholds). The bound was never violated in 105 cells. It is loose: with a fixed threshold $\rho_{\min}/2$, it reaches 0.9 at about 3–13 times the n at which the simulated probability does, and much later with Bonferroni thresholds, where the Chernoff step loses the Mills-ratio factor. The empirical $n\,\mathrm{Var}(\hat{r})/W$ lies in $[0.93, 1.11]$. With the nominal test, $P(\hat{R} = R)$ plateaus at 0.94, 0.22, 0.03 and 0.27, as Proposition 11 predicts (predicted vs simulated false-positive rates, e.g. 0.33 vs 0.32–0.33 at $\lambda = 0.9$). **F2** ($\ell \in \{0.02, 0.1, 0.3, 0.64\}$): the asymptotic detection curve matches the simulated fraction of declared non-relatives within 0.02 for $n \geq 200$; predicted $n_{1/2} = 1{,}079$ for $\ell = 0.1$, simulated 0.479 at $n = 1{,}000$. **F3**: predicted $n_{\mathrm{crit}} = 472$ ($\ell = 0.64$, $r = 0.9$) gives a simulated declared fraction 0.495 at $n = 471$; over 38 cells the asymptotic prediction is within 0.03. With travel-time lags $(0, 1, 2, 4, 8)$, $\phi = 0.9$, $r = 1$, the declared fraction still rises to 0.63 at $n = 10{,}000$, as the multi-filter part of Proposition 13 predicts.

F. Lean Formalisation: Map and Findings

Lean 4.35.0 and Mathlib v4.35.0. The build log of a clean rebuild, the `#print axioms` output for each of the 133 main theorems (all `[propext, Classical.choice, Quot.sound]`) and a grep showing no `sorry`/`admit`/`axiom` are in the supplementary material.

statement	Lean names	status
Lemmas 14 and 15	unrolled_reach_iff, unrolled_reach_iff_dist, roots_unrolled_iff, desc_root_unrolled_iff, isRel_unrolled_iff	checked
Theorem 1	edge_score, edge_score_dist, strictCond_iff_dist	checked
Proposition 4	rho_unrolled_eq, mT_le_iff, rho_unrolled_large_t	checked
Theorem 2	sRho_unrolled_exact, sRho_unrolled_bounds, tendsto_sRho_unrolled, tendsto_sRho_GT	checked
Corollary 3	tendsto_half_iff, crossEdges_eq_empty_tfae, sRho_unrolled_ge_of_cross, reisach_theorem4, limitValue_dag	checked
Theorem 5	rhoBurn_eq, sBurn_eq, rhoZ_eq, sZ_eq, sOf_window, relSet_eq_iff_src, mem_upEdges_iff	checked
Corollary 6	sBurn_eq_half_iff, sBurn_eq_half_iff_common_anc, sZ_eq_half_iff, upEdges_eq_empty_tfae, stat_half_of_strong, stat_half_of_outForest, inForest_isRel_iff, inForest_up_iff, statValue_inForest, statValue_inForest_gt_iff	checked
Table 1	collider_limit, fork_limit, chain3_limit, collider_stat, fork_stat, chain3_stat	checked
self-loop remark	edgeLoops_limit, edgeNoLoops_sRho	checked
Theorem 22 (ancestors, relatives, edge value)	isRel_unrolled_iff_NL, reduce_ancestor, relSet_unrolled_eq_iff_NL, edge_score_NL	checked
Corollary 23	sRho_unrolled_eq_NL, sRho_unrolled_ge_half_NL	checked
Proposition 26 (witness $u = j$, fresh parent)	strictNL_of_closed_walk, strictNL_of_fresh_parent, sRho_unrolled_eq_one_NL	checked
loop-free examples (Table 9)	colliderNL_sRho, edgeLoopHead_sRho, edgeNoLoops_sRho	checked
eventual periodicity, limit, same-SCC edges, Theorem 28	–	paper / enumerated
Proposition 7	–	sketch + numerics
Proposition 29 (oracle sort accuracy; optimality)	accRule_sortRule, accRule_sortRule_rho, accRule_le_sOf, accRule_le_sRho, edgeAcc_sortRule_strict, rho_not_gt_of_edge	checked
Proposition 8	–	paper only
Theorem 9(a)	fejer_identity (counting identity)	partial
Theorem 9(b)	longrun_factor, longrun_inv, selfloop_sum_lt_one	partial
Theorem 9(c,d)	–	paper only
Lemma 17	phi_le, chernoff_logmgf_le, chernoff_choice, sqrt_two_v_x	algebra only
Lemma 18	quad_M, trace_MC, lemma12_trace, lemma12_fejer, avg_toeplitz_le_tsum, lemma12_le_B, lemma12_diag, lemma12_diag_le_D	algebra only
Lemma 19	centred_cov_eq, centred_dev_offdiag, centred_dev_diag, lemma13_eps, lemma13_count	deterministic part
Lemma 20	corr_perturb	checked
Theorem 10	decide_eq_target, decide_eq_support, thm15_det, explicit_y, sOf_cntDecl_congr	deterministic part
Proposition 11	–	asymptotic
Proposition 12	prop21_corr, prop21_corr_std, sOf_saturated, sOf_const, abs_sOf_sub_half_le, thm22_const, ncrit_iff, ncrit_eq_iff	deterministic part
Proposition 13	pcorr_eq_pcov, pcorr_eq_corr_resid, prop31, prop31_std, cor32, prop33, prop33_abs, absPi_lower, absPi_upper, absPi_le_kappa, absPi_lt_kappa, absPi_eq_zero_iff, absPi_strictAntiOn, absPi_equal, rcrit_iff, rcrit_eq_iff, rcrit_example, pc_lipschitz, thm35_bracket, prop36_pcov, prop36_zero_iff, prop36_rank_one, prop36_ar1_pcov, prop36_ar1_pvar, prop36_ar1, prop36_ar1_ne_zero, prop37_pcov, prop37_bound, prop38_ratio, prop38_loading	population part checked

Modelling choices. Summary graphs are relations on a finite type with a self-loop hypothesis; anc and desc are reflexive-transitive closures ("a path exists iff a walk exists" is the one convention adopted without proof); G_T with $T = n + 1$ slices lives on $V \times \mathrm{Fin}(n + 1)$; source SCCs are represented by their member nodes (equality of SCC sets as equality of node unions is elementary and not formalised). In the finite-sample files, zero-mean variables are vectors of an arbitrary real inner-product space and lag-0 covariance is the inner product; independence enters only as vanishing inner products.

Findings. No statement was false. Formalisation added four missing nondegeneracy hypotheses to our earlier finite-sample statements: (i) the strict claims of Proposition 13 need $b_u b_v \neq 0$; (ii) $\ell_u = \rho_{uZ}^2 / r$ needs $r > 0$; (iii) nonvanishing of the multi-filter residual needs $b_u b_v \neq 0$; (iv) the partial correlation needs positive partial variances ($\mathrm{Var} Y_u > 0$ suffices). It also showed that one hypothesis of Proposition 4 and one of Lemma 20 are redundant, and that the in-forest statement needs acyclicity, not only out-degree at most one.

G. Real-Data Details

CausalRivers (Stein et al., 2025). Study 1 (P1–P3): the benchmark's connected-subgraph sampler, seeds 7000–7029, East Germany and Bavaria, $d \in \{5, 10\}$, 120 subgraphs, 791 induced edges; block means at 12 h and 24 h; nominal Fisher-z. Study 2 (P4–P6): seeds 8000–8029, 120 subgraphs, 1,376 non-relative pairs; proxy Z is the mean of standardised stations of the region outside the subgraph and outside 2 undirected river hops (429–615 stations); primary test on partial correlations with a residual Bartlett n_{eff} ($K = 100$ steps at 12 h, 50 at 24 h); median n_{eff} 139/113 (raw) and 231/211 (residual). Before conditioning, 0.98 (nominal) and 0.87 (effective) of non-relative pairs tested dependent. Median proxy contamination by river relatives was 2.6% (maximum 72%), a case of the contaminated-proxy term in Proposition 13. Post hoc and without verdict: conditioning on $k = 1, 3, 5, 10$ principal components of the leave-out stations gave dependent fractions $0.745, 0.605, 0.576, 0.491$ (12 h), with accuracy 0.580 and Spearman 0.16 at $k = 10$.

Causal Chambers (Gamella et al., 2025). Candidate selection (recorded before analysis) compared Causal Chambers, CausalRivers with measured precipitation (rejected: the driver reaches each gauge through its own catchment filter, so by Proposition 13 a scalar proxy cannot remove it), a semi-synthetic cross-sectional manufacturing benchmark (rejected: not real, not time series) and fMRI with physiological regressors (rejected: no ground truth). Wind tunnel: runs `actuators_random_walk_2-16` (run 1 excluded before analysis: truncated); 16 random variables, 26 edges, 63 non-relative pairs per run; block means over 10 rows, first differences, standardisation, $n = 999$; effective-n Fisher-z with Bartlett $K = 20$. Light tunnel: `actuators_white`, 20 variables, 39 edges, ten segments of 2,000 steps, levels. Predicted values come from the package's published ground-truth graph with constant parameters removed. Secondary analyses (no verdicts): nominal n gives slightly more false positives (0.048), as Proposition 11 predicts; levels instead of increments gave 0.054, contradicting our own secondary prediction that levels of random walks would produce many spurious dependences; without block means ($D = 1$) relative recovery drops to 0.18 and Spearman to 0.10. In the light tunnel, polariser angles act on two sensors through Malus' law, a nonlinearity with near-zero linear correlation; as pre-registered, these relative pairs were never detected, without affecting orientation.

H. Pre-Registration Timeline and Deviations

Times are commit timestamps relative to the first pre-registration commit (T_0); all events occurred on one day. The plan 4 commit was made about 3 min 36 s before its results commit, which includes a run of about two minutes; as before, the process is short and agent-driven. Commit hashes are withheld for anonymity and will be provided with the de-anonymised version. Commit timestamps are set by the committing client; the hosting service's push records are the stronger evidence and can be supplied to the area chair on request.

time	event
$T_0 + 0{:}00$	plan 1 committed: targets T1–T3, predictions P1–P3, kill rule
$T_0 + 0{:}14$	theory of Sections 3 and 4, enumeration checks, operationalisation of P1–P3 (before data)
$T_0 + 0{:}17$	study 1 results (P2, P3 fail; kill rule triggered)
$T_0 + 1{:}05$	plan 2 committed: F1–F3 targets, Lean target, predictions P4–P6, kill rule
$T_0 + 1{:}18$–$1{:}31$	Lean formalisation of Section 3 and algebraic parts of Section 4
$T_0 + 1{:}38$	finite-sample theory, simulations, operationalisation of P4–P6 (before data)
$T_0 + 1{:}44$	study 2 results (P4, P6 fail; kill rule triggered)
$T_0 + 4{:}58$	plan 3 committed: candidate scoping, graph-only predictions, P7–P9 (before any dependence statistic)
$T_0 + 5{:}02$	study 3 results (P7, P8 pass; P9 fails; replication passes)
$T_0 + 5{:}14$–$5{:}24$	Lean formalisation of the finite-sample lemmas; errata
$T_0 + 5{:}45$	manuscript draft committed (with placeholders for E1/E2)
$T_0 + 5{:}50$	plan 4 committed: E1 (test of the theorem against eight naive predictors), E2 (lagged relatives), theory addendum A1–A3 and B1–B3, graph-only predictions, design check; development runs on the study-3 data only; no dependence statistic of any held-out dataset computed
$T_0 + 5{:}54$	E1/E2 results committed as pre-registered (E1 passes 8/8 on the primary graph; Q7, Q8, Q10 pass, Q9 fails); out-of-sample F checks; run logs

Deviations (all recorded at the time). (1) Plan 1 said "theorem-predicted s_ρ" without naming the version; the stationary version was fixed in writing before data were loaded (the finite version would predict $> 1/2$ for every subgraph, including chains). (2) Self-loops are excluded from the predicted average because the estimator cannot orient them; this does not change which side of $1/2$ the prediction is on. (3) Only 6 of 120 study-1 subgraphs lacked a confluence; the plan set no minimum size for that group, and its trivial pass is not interpreted. (4) Plan 2 did not say which test defines "dependent" in P4; the effective-n test was fixed as primary before data, and the nominal test fails P4 as well. (5) In study 3, a planned triad-reliability check was not applicable (the measured driver reaches only four mutually related barometers) and is not interpreted. (6) The P9 failure is close to the threshold and its CI contains 0.3; as pre-registered, the point estimate decides. (7) Plan 4 (E1/E2): no threshold, unit, seed, K, L, D, graph or node set changed after the results were seen. A stray cache file committed with plan 4 was removed in the results commit (no content change). The F-check tolerance of ± 0.03 is the one fixed in plan 4; the citris "predicted accuracy" is undefined (no full-set edge of G_{lin} is predicted 1). The constant-$1/2$ comparison is post hoc. The choice of G_{lin} as the primary graph was pre-registered, together with the secondary analysis on the published graph G_D, which is the one on which the theorem fails against the sensor indicator on strictly ordered edges.

I. Experiment E1: Discriminating Test Against Naive Predictors

Status and provenance. Pre-registered in plan 4 (Appendix H); the plan was committed before any dependence statistic of any held-out dataset was computed. Done before it: file hashes, column and row counts, regeneration of the `citris_1` input protocol from its published generator (it matches the recorded inputs exactly), graph-only predictions, and development runs on the study-3 data only. All numbers below come from the committed result files of the experiment folder (`results4.json`, `summary4.txt`); none is estimated or rounded beyond the printed digits.

Why `citris_1`. Every published Causal Chambers graph is bipartite except the pressure-control configuration, whose only experiment varies no independent input, so we rejected it. The light-tunnel `citris_1` series (100,000 steps about 1.8 s apart) is generated by a time-series structural model among the actuators: red is a reflected random walk, green moves halfway towards red and blue a quarter of the way towards green (each plus a uniform step), pol_1 is a random walk and pol_2 moves towards or away from pol_1 according to $\text{sign}(\text{blue} - \text{red})$. With the sensors this is a chain of length 3, red $\to$ green $\to$ blue $\to$ sensors. The secondary data are snapshot (i.i.d.) structural models among the actuators (`buchholz_1_obs`; `scm_{1,2,4,5}_reference`, including a chain and a collider) and the wind-tunnel regime-jump runs (bipartite; disagreements arise only from latent-node ties).

Linear-zero lemma (the refinement). Write $\mathcal{C}$ for the colour block (red, green, blue, ir_1, vis_1, ir_2, vis_2, ir_3, vis_3, current) and $\mathcal{Q}$ for the polariser block (pol_1, pol_2, angle_1, angle_2). *B1 (proved for the actuators).* Conditionally on the flag and the colour paths, the law of the polariser paths is invariant under negation, because every update (uniform draw on a symmetric interval, move towards or away with a sign fixed by the colours, reflection at ± 90, rounding to 0.1) commutes with negation. *B2 (assumed).* The first-two-layer sensors depend only on the colours and independent noise; ir_3 and vis_3 depend on the polarisers only through an even function of $\theta_1 - \theta_2$ (Malus' law with aligned zero marks); each angle sensor is affine in its polariser plus independent noise. This is a physical assumption that the design documents do not let us check. *B3.* Under

B1 and B2 every variable of $\mathcal{C}$ has zero cross-covariance, at every pair of times, with every variable of $\mathcal{Q}$ (a product of an even and an odd function of the polariser path has conditional mean zero). The six real edges red $\to$ pol_2, blue $\to$ pol_2 and $\text{pol}_k \to \text{ir}_3$, vis_3 therefore carry no linear dependence. The relatives that a correlation-based estimator targets are those of the *linear-dependence graph* G_{lin} (the design graph G_D without these six edges), which has two weak components, each with a single source (red; pol_1). By Corollary 6, every edge inside a block has predicted value $1/2$ for every node set; the six cross-block edges are compared by block size: red $\to$ pol_2 and blue $\to$ pol_2 are predicted 0 (anti-sorted) and $\text{pol}_k \to \text{ir}_3$, vis_3 are predicted 1 for the full set. Faithfulness inside the blocks is assumed, as in Proposition 8. A zero-mark offset or a non-affine angle sensor would break B2 and move the prediction towards G_D.

Metric and tests. For each edge instance (the full node set plus 20 subgraphs per unit) the observed value is $o \in \{0, \frac{1}{2}, 1\}$ from the estimated counts, and the loss is $|o - p|$ for the predicted value p. For each competitor take the instances where it disagrees with the theorem; $\Delta = \text{loss(theory)} - \text{loss(competitor)}$, pooled, with a unit-level exact sign-flip test (1,024 flips for 10 units) and Holm correction over eight competitors, and a unit-bootstrap CI. E1a (prerequisites): non-relatives declared < 0.5 and relatives detected ≥ 0.8. E1b: $\Delta \leq -0.10$ and Holm $p \leq 0.05$ for all eight. E1c: the same restricted to instances whose own pair is declared dependent ($\Delta < 0$). Separation was to be claimed iff E1a, E1b and E1c all pass on `citris_1`. Kill rules: K1 (E1a fails: no claim), K2 (a competitor is significantly better: no claim against it), K3 (E1b passes and E1c fails: attributed to power-induced ties). None was triggered on the primary analysis.

Table 4. E1 primary, `citris_1`, graph G_{lin}, lag 0, 1,662 instances. Δ [95% CI]; all rows: 10/10 units, Holm $p = 0.0078$. "Exact": share of instances whose predicted value equals the observed one. "Spearman": agreement of the mean predicted edge value with observed accuracy over 200 subgraphs.

predictor	Δ [CI]	Δ, detected edges	exact	Spearman
theorem	—	—	0.992	0.962
sensor indicator	-0.502 [-0.507, -0.496]	-0.500	0.203	0.408
ancestors, full graph	-0.515 [-0.519, -0.510]	-0.500	0.147	0.334
ancestors, subgraph	-0.509 [-0.513, -0.505]	-0.500	0.120	0.121
in-degree, full graph	-0.515 [-0.520, -0.511]	-0.500	0.180	0.349
in-degree, subgraph	-0.510 [-0.514, -0.505]	-0.500	0.143	0.105
depth, full graph	-0.515 [-0.519, -0.510]	-0.500	0.147	0.334
depth, subgraph	-0.510 [-0.514, -0.505]	-0.500	0.114	0.113
every edge 1	-0.533 [-0.536, -0.530]	-0.500	0.115	undefined

Observed values given the theorem's prediction: predicted $1/2$ (1,365 instances) observed exactly $1/2$ in 99.2%; predicted 1 (187 instances, polariser $\to$ ir_3, vis_3) observed 1 in 99.5%; predicted 0 (110 instances, red and blue $\to$ pol_2) observed 0 in 99.1%. The detected-edge subset has 1,338 instances, all observed exactly $1/2$. Prerequisites: non-relatives declared 0.000 [0, 0] (40 pairs per unit), relatives detected 0.969 [0.945, 0.992] (51 pairs per unit).

Strict subset and the power safeguard. On the 297 instances where the theorem predicts 0 or 1 it beats all eight ($\Delta = -0.539$ for the sensor indicator, -0.948 for ancestors and depth on the full graph, -0.991 for "every edge 1"; Holm $p = 0.0078$, 10/10 units). There, an estimator without power would favour the competitors (the "all counts tied" baseline is $+0.443$ to 0), so this part cannot come from power-induced ties.

Published graph G_D (secondary). Applying the theorem to the unmodified design graph, relatives detection is 0.659 [0.643, 0.675] (75 relative pairs per unit) and non-relatives declared 0.000. Overall the theorem still beats all eight ($\Delta = -0.323$ for the sensor indicator; -0.424 for ancestors, depth and "every edge 1"; -0.420 and -0.417 for in-degree; 10/10 units, Holm $p = 0.0078$), and its exact-match rate is 0.714. But on the 442 edges where G_D predicts strict order (1), the theorem matches 0.222 and the sensor indicator 0.319; the sensor indicator is significantly better ($\Delta = +0.453$ [0.410, 0.492], theorem better in 0/10 units, reverse Holm $p = 0.002$). A constant-$1/2$ rule (post hoc) matches 0.815 of all instances, against 0.714 for the theorem on G_D and 0.992 on G_{lin}; mean $|o - 1/2| = 0.092$ against 0.166 for the theorem on G_D. The Spearman statistic is 0.341 (theorem on G_D) against 0.408 (sensor indicator). This is why the claim is restricted to the linearly dependent relatives.

Lagged estimator on `citris_1` (secondary). With $L = 5$ and G_{lin}: detection 0.976 [0.953, 0.992], non-relatives declared 0.000, 8/8 passed ($\Delta = -0.505$ to -0.535), theorem exact match 0.998, Spearman 0.996.

On the snapshot sets, relatives detected: 0.933 (`buchholz_1_obs`), 1.000 (`scm_1`), 0.965 (`scm_2`), 1.000 (`scm_4`, `scm_5`); non-relatives declared 0.000 wherever non-relative pairs exist (`scm_4` has none). The chain (`scm_4`) gives $1/2$ on all

Table 5. E1 on i.i.d. structural-model sets (no verdict; secondary). Competitors' exact-match range over the eight predictors. Units: `scm_1` 5, the others 10 each (45 pooled).

set	instances	theorem exact	competitors exact	Δ range	result
`buchholz_1_obs`	1,712	0.791	0.328–0.375	-0.499 to -0.496	pass 8/8
`scm_1_reference`	867	1.000	0.314–0.392	-0.500	not met[†]
`scm_2_reference`	1,650	0.812	0.324–0.408	-0.443 to -0.426	pass 8/8
`scm_4_reference` (chain)	2,267	1.000	0.000–0.088	-0.500	pass 8/8
`scm_5_reference` (collider)	2,238	1.000	0.464–0.495	-0.500	pass 8/8
pooled (45 units)	8,734	0.923	0.283–0.322	-0.489 to -0.487	pass 8/8, Holm $p = 8 \times 10^{-5}$

[†]Five units: the smallest attainable Holm-adjusted p is $8/32 = 0.25$.

2,267 instances (observed exactly $1/2$ every time); the collider gives 1 on edges into the collider and its sensors and $1/2$ elsewhere (all 2,238 instances exact). Pooled, the theorem's strict predictions number 2,768 (observed value 1 on 2,403, $1/2$ on 349, 0 on 16: exact 0.868); on them only the sensor indicator disagrees often (390 instances, $\Delta = -0.5$, 35/35 units) and the other competitors coincide with the theorem. The pooled Spearman is 0.913 for the theorem and -0.138 for the sensor indicator. Post hoc, a constant-$1/2$ rule matches 0.688 of the pooled instances against 0.923.

Wind-tunnel regimes (tertiary, 64 units, no verdict). The same test passes against all eight in each configuration. $D = 10$, $L = 0$: non-relatives declared 0.014, relatives detected 0.592, $\Delta = -0.203$ $[-0.227, -0.179]$ in 63/64 units, exact match 0.539 against 0.432; detected-edge subset $\Delta = -0.169$. $D = 1$, $L = 30$: 0.003, 0.398, $\Delta = -0.225$ $[-0.250, -0.197]$ in 62/64 units, exact 0.443 against 0.353; detected-edge subset $\Delta = -0.134$. $D = 1$, $L = 0$: $\Delta = -0.218$ (64/64 units), exact 0.411 against 0.326; $D = 10$, $L = 3$: $\Delta = -0.218$, exact 0.533 against 0.411. Detection of 0.33–0.59 means that part of the advantage is power-induced, as plan 4 stated in advance. Post hoc, the constant-$1/2$ rule has mean absolute error 0.311 against 0.363 for the theorem at $D = 1$, $L = 30$ (and 0.313 against 0.294 at $D = 10$, $L = 0$), and matches 0.378 (0.375) of the instances against 0.443 (0.539).

What E1 does and does not show. It shows that, on one multi-layer light-tunnel series, the estimated relatives counts are tied along the whole chain red $\to$ green $\to$ blue $\to$ sensors and between the polariser block and the angle sensors, and strictly ordered (including against the causal direction) on red, blue $\to \mathrm{pol}_2$ — and that the source-component rule of Theorem 5 and Corollary 6 predicts this while eight naive rules do not. It does not show that the theorem predicts relatives under the published graph; the faithfulness of the graph to linear dependence fails here by design. Most of the separation is in the ties, which naive rules cannot express. The evidence is one device in a favourable regime, and the wind tunnel, where detection is limited by F1, gives weaker support.

J. Experiment E2: Lagged Relatives

Theory (plan 4, addendum A). For $L \in \{0, 1, \dots\}$ let the lag-L relatives of u be u and every v with $\gamma_{uv}(h) \neq 0$ for some $|h| \leq L$. *A1 (proved).* Under the independent-innovations assumption behind Proposition 8, $\mathrm{rel}_0 \subseteq \mathrm{rel}_L \subseteq \mathrm{rel}_\infty \subseteq \mathrm{rel}_S$; for a VAR model all are equal for almost every parameter value; and Theorem 5, Corollary 6 hold verbatim with "relatives" read as rel_L under lag-L faithfulness. Lag-0 faithfulness implies lag-L faithfulness; the converse fails (a white-noise source s with an edge $s \to j$ and no self-loop at s has $\gamma_{sj}(0) = 0 \neq \gamma_{js}(1)$). *Estimator.* Declare u, v relatives iff $\max_{|h| \leq L} |\operatorname{atanh} \hat{r}_{uv}(h)| \sqrt{\hat{n}_{\mathrm{eff},uv}(h) - 3} > z_c(L)$, with $\hat{n}_{\mathrm{eff},uv}(h) = (n - |h|)/\hat{B}_{uv}$ and $z_c(L) = \Phi^{-1}(1 - \alpha/(2m(2L + 1)))$, a Bonferroni correction over the m pairs and the $2L + 1$ lags; $L = 0$ is exactly the study-3 test. *A2 (proved for finite n).* For independent stationary Gaussian a, b the null variance of the lag-h cross-covariance has the same bound at every h, so one Bartlett factor serves all lags. *A3 (deterministic part proved, constants sketched).* The recovery bound of F1 holds with $\rho_{\min}$ replaced by $\rho^*_{\min,L} = \min_{\text{relative pairs}} \max_{|h| \leq L} |\rho_{ab}(h)| \geq \rho_{\min}$ (the gain) and a factor $2L + 1$ in the failure probability and the Bonferroni count (the cost): for 105 pairs at $\alpha = 0.01$, z_c rises from 3.90 ($L = 0$) to 4.35 ($L = 3$) and 4.80 ($L = 30$). A lagged correlation must therefore exceed the lag-0 correlation by about 1.12 ($L = 3$) or 1.23 ($L = 30$) times to help.

Design. Held-out data: the wind-tunnel regime-jump runs (`regime_jumps_single`, `regime_jumps_multi`), $2 \times 32 = 64$ regimes of 10,000 steps, in the same archive as the study-3 data but never analysed in study 3. Load_in, load_out, pot_1, pot_2 follow independent mirrored random walks within a regime; the hatch and sensor parameters are constant within a regime. The first 100 rows of each regime are dropped. Graph: 20 edges without the hatch (19 predicted 1, $\mathrm{pot}_1 \to \mathrm{signal}_1$

predicted $1/2$); 54 non-relative and 51 relative pairs; 1,280 subgraphs (10 of size 5 and 10 of size 8 per unit). Primary: $D = 1$ (first differences), $L = 30$ (about 9 s) against $L = 0$; secondary: block means with $D = 10$ and $L = 3$ against $L = 0$. Success required Q7–Q10 all to pass at the primary configuration. Kill rules: Q7 fails (saturation) or Q10 fails (lags do not help detection, and the lagged estimator is not recommended). Plan 4 states that E2 is not a re-analysis of P9 and that P9's verdict is final.

Table 6. E2, 64 held-out regimes, 95% unit-bootstrap CIs. Q8: accuracy on the 1,216 instances predicted 1. Q10: relatives detected, lagged minus lag 0 (non-relatives declared difference in the last column).

configuration	Q7	Q8	Q9	Q10	non-rel. diff.
primary D=1, L=30	0.003 [0.001, 0.005] pass	0.553 [0.527, 0.579] pass	0.243 [0.193, 0.294] **fail**	$+0.069$ [0.060, 0.079] pass	$+0.001$ [-0.001, 0.002]
secondary D=10, L=3	0.011 [0.008, 0.014]	0.711 [0.692, 0.729]	0.202 [0.152, 0.253] fail	-0.007 [-0.012, -0.003] fail	-0.002 [-0.004, -0.0003]
lag 0, D=10 (replication of study 3)	0.014 [0.010, 0.018]	0.742 [0.722, 0.760]	0.202 [0.145, 0.259] fail	—	—
lag 0, D=1	0.002 [0.001, 0.004]	0.552 [0.532, 0.570]	0.120 [0.066, 0.176] fail	—	—

Detection of the 51 relative pairs: $D = 1$: 0.328 [0.312, 0.345] at lag 0 and 0.398 [0.383, 0.412] with $L = 30$; $D = 10$: 0.592 [0.584, 0.599] and 0.585 [0.576, 0.592] with $L = 3$ (lags reduce detection: the Bonferroni cost exceeds the gain). On `citris_1` with $L = 5$ against $L = 0$: 0.976 against 0.969, difference $+0.008$ [0, 0.024], so Q10 fails there; detection is already nearly saturated at lag 0. **Verdict.** Q9 fails at the primary configuration, so E2 does not succeed; no kill rule is triggered (Q7 and Q10 pass). Lags raise detection at fine resolution without adding false positives, as A1 and A3 predict. The study-3 estimator (lag 0) rerun on these held-out regimes fails the graded criterion again (0.202 at $D = 10$, 0.120 at $D = 1$): the P9-type failure repeats on held-out data. P9's original verdict (0.267, failed) stands; E2 is not presented as rescuing it. The development check of plan 4 showed that, on the study-3 data with the same estimator, drawing different subgraph seeds moves the P9 statistic from 0.267 to 0.338. We report this as variability of the statistic (about 0.07 from the subgraph draw alone); it is a development run, not a verdict, and it does not change the P9 outcome.

Table 7. Detection rate per edge over 64 regimes: share of regimes in which the pair was declared dependent.

edge	D=1, L=0	D=1, L=30	D=10, L=0
load_in → rpm_in	0.859	0.875	1.000
load_in → rpm_out	0.000	0.172	0.094
load_in → current_in	1.000	1.000	1.000
load_in → current_out	0.000	0.000	0.047
load_out → rpm_in	0.000	0.016	0.000
load_out → rpm_out	0.844	0.969	1.000
load_out → current_out	1.000	1.000	1.000
load_out → current_in	0.000	0.000	0.000
load_in → pressure_intake	0.000	0.094	0.953
load_out → pressure_intake	0.000	0.031	0.906
load_in → pressure_upwind	0.562	0.984	1.000
load_out → pressure_upwind	0.016	0.812	1.000
load_in → pressure_downwind	0.734	0.953	1.000
load_out → pressure_downwind	0.750	1.000	1.000
$\text{pot}_1 \to \text{signal}_1$	0.359	0.156	0.969
$\text{pot}_1 \to \text{signal}_2$	0.266	0.109	0.969
$\text{pot}_2 \to \text{signal}_2$	0.000	0.000	0.266
pot_1, load_in, load_out → mic	0.000	0.000	0.016 (pot_1), 0.000

Lags recover the pressure edges at fine resolution (load_in → upwind pressure $0.562 \to 0.984$, load_out → upwind pressure $0.016 \to 0.812$) but lose the pot edges to the larger threshold ($0.359 \to 0.156$ and $0.266 \to 0.109$); every microphone edge, $\text{pot}_2 \to \text{signal}_2$ and the cross-current edges stay undetected. Even at its best ($D = 10$, lag 0) wind-tunnel detection (0.592) is far below that of the light tunnel, so lags are not the main cause of the F1 shortfall.

All six checks pass. With lags the observed detection sits at or slightly below $\bar{p}_{\max}$, consistent with strongly correlated lag statistics; the independence formula overstates both detection and false positives, as the theory said it would. For `citris_1` the predicted accuracy is undefined because no full-set edge of G_{lin} is predicted 1 inside a block.

Table 8. Out-of-sample F-theory checks: the first half of each unit predicts the second half. Predicted detection is a bracket $[\bar{p}_{\max}, \bar{p}_{\mathrm{ind}}]$ when lags are used. Pass if the observed value lies within 0.03 of the bracket (criterion fixed in plan 4).

configuration	predicted detection	observed	non-rel. declared, pred. / obs.	accuracy, pred. / obs.
wind D=1, L=0	0.250	0.242	0.007 / 0.003	0.573 / 0.590
wind D=1, L=30	[0.344, 0.426]	0.332	[0.021, 0.074] / 0.003	0.504 / 0.506
wind D=10, L=0	0.548	0.533	0.014 / 0.008	0.649 / 0.632
wind D=10, L=3	[0.536, 0.554]	0.524	[0.016, 0.027] / 0.008	0.628 / 0.618
`citris_1`, L=0	0.935	0.945	0.007 / 0.000	undefined
`citris_1`, L=5	[0.940, 0.988]	0.945	[0.010, 0.045] / 0.000	undefined

K. Summary Graphs without Self-Loops

Here $G_{\mathrm{S}} = (V, E_{\mathrm{S}})$ is an arbitrary directed graph with $|E_{\mathrm{S}}| \geq 1$; no loop assumption is made. B_n is the boolean n-th power of the adjacency matrix ($B_0 = I$), $\Lambda(u, v) = \{n : B_n[u, v] = 1\}$ the set of lengths of walks from u to v, and u is *fresh* if it has no incoming edge in E_{S} (a self-loop is an incoming edge). Status: Theorem 22 and Corollary 23 are machine-checked; the rest of this section is proved on paper and checked by enumeration, as marked.

Lemma 21 (ancestors and roots without loops)**.** *$u(a) \in \mathrm{anc}(v(t))$ iff $a \leq t$ and $t - a \in \Lambda(u, v)$. A node $u(a)$ is a root of G_T iff $a = 1$ or u is fresh.*

Lean: `unrolled_reach_iff` *(already free of the loop hypothesis).*

Proof. Every edge of G_T raises time by one, so paths from $u(a)$ to $v(t)$ are the lifts of walks of length exactly $t - a$, and every walk lifts. Without loops one cannot pad. A node $u(a)$ with $a \geq 2$ has a parent iff some edge (w, u) exists. □

Theorem 22 (edge value without loop assumption)**.** *Let $(i, j) \in E_{\mathrm{S}}$ and $e_t = (i(t), j(t{+}1))$, $1 \leq t < T$. Then $s_\rho(e_t) \in \{\frac{1}{2}, 1\}$ independently of T, and $s_\rho(e_t) = 1$ iff $\mathrm{Strict}_{ij}(t)$ holds, where $\mathrm{Strict}_{ij}(t)$ is*

$$\exists u:\ t \in \Lambda(u, j),\ t - 1 \notin \Lambda(u, i) \quad \textit{or} \quad \exists \textit{ fresh } u,\ \exists n \in \{1, \ldots, t\}:\ n \in \Lambda(u, j),\ n - 1 \notin \Lambda(u, i).$$

Lean: `isRel_unrolled_iff_NL`, `reduce_ancestor`, `relSet_unrolled_eq_iff_NL`, `edge_score_NL`.

Proof. As in Theorem 1, $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$, and the score is 1 iff $\mathrm{roots}(j(t{+}1)) \setminus \mathrm{roots}(i(t)) \neq \emptyset$ (Reisach et al., 2026, Thm. 1, Lemma 1). By Lemma 21 an element of the difference is $u(a)$ with ($a = 1$ or u fresh), $n = t{+}1{-}a \in \Lambda(u, j)$ and $u(a) \notin \mathrm{roots}(i(t))$. The case $a = t + 1$ would make j fresh, impossible since j has the parent i; so $a \leq t$ and, by the lemma again, $u(a)$ is not a root of $i(t)$'s ancestry iff $n - 1 = t - a \notin \Lambda(u, i)$. The branch $a = 1$ is $n = t$ with no condition on u; the other is a fresh u with $1 \leq n \leq t$. (The Lean proof avoids Lemma 1: the extra relative is $u(0)$ in the first case and a node of a fresh u in the second, and `reduce_ancestor` pushes any common ancestor back along parents to slice 0 or to a fresh node.) □

Corollary 23. *For every summary graph and $T \geq 2$, $s_\rho(G_T) = \frac{1}{2} + \frac{1}{2(T-1)|E_{\mathrm{S}}|} \sum_{(i,j) \in E_{\mathrm{S}}} N_T(i, j)$ with $N_T(i, j) = \#\{t < T : \mathrm{Strict}_{ij}(t)\}$; in particular $s_\rho(G_T) \geq 1/2$. With a loop at every node no node is fresh, $\Lambda(u, j) = [d(u, j), \infty)$, and $\mathrm{Strict}_{ij}(t)$ becomes $\exists u : d(u, j) \leq t \leq d(u, i)$, i.e. Theorem 1.*

Lean: `sRho_unrolled_eq_NL`, `sRho_unrolled_ge_half_NL`.

Exhaustive check of Theorem 22 against brute-force relatives: $p = 2, 3$ ($T = 2, \ldots, 9$), all 65,535 graphs with $p = 4$ ($T = 3, 5, 8$), and random graphs with $p = 5, 6$; 0 mismatches in 120, 3,577, 196,605, 8,928 and 4,494 (graph, T) cases (enumerated, not part of the Lean proof).

Proposition 24 (limit exists and is rational; paper proof, enumerated)**.** *(B_n) is eventually periodic: there are $n_0, \pi \geq 1$ with $B_{n+\pi} = B_n$ for $n \geq n_0$. Hence $L = \lim_T s_\rho(G_T) = \frac{1}{2} + \frac{1}{2|E_{\mathrm{S}}|} \sum_e \delta_e$ with $\delta_e = 1$ if some fresh u has $n \geq 1$, $n \in \Lambda(u, j)$, $n - 1 \notin \Lambda(u, i)$, and otherwise δ_e the density, over one period of $t > n_0$, of t with some u, $B_t[u, j] = 1$, $B_{t-1}[u, i] = 0$; and $|s_\rho(G_T) - L| \leq (n_0 + \pi + 1)/(2(T - 1))$.*

Proof. B_{n+1} is a function of B_n and there are finitely many boolean matrices. A fresh witness is monotone in t and has its first n at most $n_0 + 1 + \pi$ (the set $\Lambda(u, j) \setminus (\Lambda(u, i) + 1)$ is eventually periodic), so such an edge is strict for all

$t \geq n_0 + 1 + \pi$. Without a witness, $\mathrm{Strict}_{ij}(t)$ depends for $t > n_0$ only on $t \bmod \pi$. Counting strict $t < T$ gives the error bound; average over edges. □

Checked on 1,115 graphs (all $15 + 511$ graphs with $p = 2, 3$, 390 random with $p = 4$, 199 random with $p = 5, 6$): the exact value at $T = 600$ is within the bound and $(L - \frac{1}{2})\, 2|E_{\mathrm{S}}|\, \pi$ is an integer in every case.

Proposition 25 (edges inside an SCC; paper proof, enumerated)**.** *Let $(i, j) \in E_{\mathrm{S}}$ with j reaching i and no fresh witness. Then $\mathrm{Strict}_{ij}(t)$ fails for every $t \geq n_0 + 1$.*

Proof. Fix u; put $f(n) = B_n[u, j]$, $h(n) = B_{n-1}[u, i]$. The edge gives $h(n) \leq f(n)$. If m is the length of a walk $j \to i$ and $c = m + 1$, then $f(n) \leq h(n + c)$. Take $k \geq 1$ with kc a multiple of π. For $n \geq n_0 + 1$, $h(n) \leq f(n) \leq h(n + c) \leq \cdots \leq h(n + kc) = h(n)$, so $f(n) = h(n)$ for every u. □

Proposition 26 (crossing edge into a cyclic node)**.** *If (i, j) is crossing and j lies on a cycle with period g (gcd of the closed-walk lengths at j), then $\mathrm{Strict}_{ij}(t)$ holds for every $t \in \Lambda(j, j)$, which contains all large multiples of g, so $\delta_e \geq 1/g$, and $\delta_e = 1$ if $g = 1$. A fresh parent $u \neq i$ of j makes every layer strict.*

Lean: `strictNL_of_closed_walk`, `strictNL_of_fresh_parent`, `sRho_unrolled_eq_one_NL`.

Proof. Take $u = j$: $t \in \Lambda(j, j)$ and $t - 1 \notin \Lambda(j, i) = \emptyset$. Closed-walk lengths at j form a numerical semigroup in $g\mathbb{Z}$, which contains all large multiples of g. □

Corollary 27 (loop-free versions of the loop-dependent statements)**.** *(a) If every weakly connected component of G_{S} is strongly connected (any loop pattern, any periods), then $L = 1/2$ and $s_\rho(G_T) - \frac{1}{2} \leq n_0/(2(T - 1))$ (paper proof from Proposition 25; Reisach et al.'s Theorem 4 without loops). (b) The converse and the closed form are false without loops (Table 9). (c) If every crossing edge has a head on a cycle, then $L > 1/2$ iff $E_\times \neq \emptyset$.*

Table 9. Loop-free examples (L: finite limit, S: stationary). Lean: `edgeNoLoops_sRho`, `colliderNL_sRho`, `edgeLoopHead_sRho`; the others are computed values (example scripts in the supplement).

summary graph	without loops	with loops
single edge $0 \to 1$	$s_\rho(G_T) = 1/2$ for all T	$L = 2/3$, $S = 1/2$
chain $0 \to 1 \to 2$	$L = 1/2$	$L = 0.7$
2-cycle $\{0, 1\}$ plus $0 \to 2$	$L = 1/2$	loop at 2 only: $L = 5/8$
collider	$L = S = 1$	$L = S = 0.7$
edge, loop at head only	$L = S = 1$	–

For an acyclic loop-free graph, $L(u, v)$ is finite, $B_n = 0$ for $n \geq p$, and $L = \frac{1}{2} + \#\{e : \text{fresh witness}\}/(2|E_{\mathrm{S}}|)$ (paper proof); for a loop-free in-forest an edge (i, j) has a fresh witness iff j has another parent, so $L = \frac{1}{2} + (m - n_1)/(2m)$ with $m = |E_{\mathrm{S}}|$ and n_1 the number of nodes with one parent, which equals the stationary value below. On 362 random acyclic loop-free graphs $L = S$ in all cases, and the in-forest formula holds on 298 in-forests.

Theorem 28 (stationary value without loops; (a), (c) proved, (b) sketched)**.** *Let G_{S} be arbitrary, $G_{\mathbb{Z}}$ the infinite-past unrolled DAG on $V \times \mathbb{Z}$, and $\Delta(v, w) = \{m - n : \exists u,\ n \in \Lambda(u, v),\ m \in \Lambda(u, w)\}$. Then $\mathrm{rel}_{G_{\mathbb{Z}}}(v(t)) = \{w(t + d) : d \in \Delta(v, w)\}$ (proved), the edge $(i(t), j(t{+}1))$ is strict iff $\Delta(j, w) + 1 \neq \Delta(i, w)$ for some w (proved; always $\Delta(i, w) \subseteq \Delta(j, w) + 1$), and $S^{\mathbb{Z}} = \frac{1}{2} + |E^{\mathbb{Z}}_{\uparrow}|/(2|E_{\mathrm{S}}|)$. (a) Outside $|d| < n_0 + \pi$, $\Delta(v, w)$ is π-periodic, so $E^{\mathbb{Z}}_{\uparrow}$ is detected with witnesses of $|d| \leq n_0 + 2\pi$ (proved). (b) In a finite window after a burn-in $B \geq 2n_0 + 3\pi + 1$ the windowed value equals $S^{\mathbb{Z}} + O(K_0/T)$, $K_0 = n_0 + 2\pi$ (*sketched*). (c) $S^{\mathbb{Z}} = 1/2$ whenever every weak component is strongly connected, whatever its periods (a node colouring lab with $\Delta(v, w) = \mathrm{lab}(w) - \mathrm{lab}(v) + g\mathbb{Z}$) (proved). With loops everywhere Δ is $\mathbb{Z}$ or empty and this is Theorem 5.*

Brute-force relatives in G_{B+T} ($B = 40$, $T = 90$, edges at distance ≥ 30 from the window ends) agree with $E^{\mathbb{Z}}_{\uparrow}$ on every edge of all $15 + 511$ graphs with $p = 2, 3$ and 248 random graphs with $p = 4$ (enumerated). Not formalised in Lean: periodicity, Proposition 25, Corollary 27(a), Theorem 28.

L. Random Summary Graphs: Derivations and Checks

All graphs here are a non-loop random graph plus a self-loop at every node, so $|E_{\mathrm{S}}| = p+m$ with m the number of non-loop edges, and for acyclic graphs $|E_{\times}| = m$.

In-forests (proved identities). Let the non-loop part be an in-forest with m edges and n_1 nodes with exactly one parent. By Corollaries 3 and 6, $L = \frac{1}{2}+\frac{m}{2(m+p)}$, $S = \frac{1}{2}+\frac{m-n_1}{2(m+p)}$, $L-S = \frac{n_1}{2(m+p)}$ (the last checked to machine precision on 18 random forests). For a sequence of in-forests with parent-count distribution π and mean μ: $L \to \frac{1}{2} + \frac{\mu}{2(1+\mu)}$ and $S \to \frac{1}{2} + \frac{\mu-\pi_1}{2(1+\mu)}$. Exact computation at $p = 20{,}000$ (6 seeds each): uniform recursive in-tree ($\pi_1 = 1/4$, $S \to 11/16 = 0.6875$, measured 0.68739 ± 0.00026); preferential attachment (measured $\pi_1 = 0.1656$, $S = 0.70859 \pm 0.00028$); uniform random labelled tree rooted and oriented to the root (Poisson(1) parents, $\pi_1 \to e^{-1}$, $S \to \frac{1}{2} + (1-e^{-1})/4 = 0.65803$, measured 0.65783 ± 0.00020). In all three $L = 0.74999$.

Directed ER, finite limit (sketch). $G(p, c/p)$, $m/p \to c$, so it suffices to show that the fraction of non-loop edges inside an SCC tends to x^2. Around a uniformly chosen non-loop edge (i, j) the local limit is: i has an independent Galton–Watson Poisson(c) tree of ancestors, j an independent one of descendants, with further independent Poisson(c) parents and children. A node lies in the giant SCC iff both its ancestor and descendant trees are infinite (the product form for a factorising degree law is stated in Dorogovtsev et al., 2001; we checked only its abstract). Since $i \to j$, both endpoints are in the giant SCC iff the ancestor tree of i and the descendant tree of j are infinite: probability x^2. Outside the giant the graph is subcritical in the dual sense and non-trivial SCCs carry $o(p)$ edges. The two analytic inputs are classical; *the concentration step (second-moment bound for the edge counts) is not written out.*

Directed ER, stationary value (sketch). Source SCCs are, up to $O(1)$, the nodes without non-loop parents, and $(i, j) \in E_{\uparrow}$ iff some other parent k of j has a source ancestor not reaching i. In the local limit, a source s reaches i iff s has an infinite descendant tree and i an infinite ancestor tree (a tree path would use the edge $i \to j$ backwards). If the ancestor tree of i is finite (probability $1-x$) any other parent works: contribution $(1-x)(1-e^{-c})$. Otherwise a witness needs a source with finite descendant tree; let d be the probability that a node has in-degree 0 or a parent all of whose other children have finite descendant trees (each with probability $1-x$) and which again satisfies the property. It solves $1-d = e^{-c(1-x)d} - e^{-c}$ and is the smallest fixed point because the event is an increasing union of depth-truncated events. The number of qualifying other parents is Poisson$(c(1-x)d)$, giving $x(1-x)(1-e^{-c(1-x)d})$. *This fixed point is derived heuristically and not proved.* Large c: $1-x \sim e^{-c}$, $L - \frac{1}{2} \sim e^{-c}$, $S - \frac{1}{2} \sim e^{-c}/2$.

Layered DAG (sketch; L proved given edge concentration). M layers of width w, each node of layer $l+1$ receiving an edge from each node of layer l with probability c/w; as $w \to \infty$ with $\mu = c(M-1)/M$, $L \to \frac{1}{2} + \frac{\mu}{2(1+\mu)}$ and $S \to \frac{1}{2} + \frac{\mu(1-e^{-c})}{2(1+\mu)}$. For fixed w and deep M the stationary value falls below this (observed 0.7824 against 0.7878 at $w = 50$, $M = 400$, $c = 2$); this correction is not derived.

Table 10. Exact computation (means ± s.d. over seeds; SCCs by scipy and source sets by bitsets) against Proposition 7.

c	p	L exact	L formula	S exact	S formula
0.5	20000	0.6666 ± 0.0011	0.6667	0.5658 ± 0.0014	0.5656
1.0	20000	0.7486 ± 0.0008	0.7500	0.6557 ± 0.0013	0.6580
1.5	50000	0.6975 ± 0.0020	0.6981	0.6159 ± 0.0018	0.6163
2.0	50000	0.6214 ± 0.0022	0.6217	0.5629 ± 0.0014	0.5632
3.0	100000	0.5434 ± 0.0010	0.5433	0.5215 ± 0.0006	0.5214
5.0	20000	0.5059 ± 0.0005	0.5058	0.5028 ± 0.0003	0.5029

L at $c = 1$ converges as p grows: 0.7466 ($p = 2000$), 0.7486 ($2 \cdot 10^4$), 0.74976 ($2 \cdot 10^5$), 0.74992 (10^6); at $c = 2$, 0.62162 ± 0.00023 at $p = 10^6$ against 0.62170. The giant-SCC fraction matches x^2 (0.6353 against 0.6349 at $c = 2$). The second fixed point matters: with $d = 1$ the formula would give $S = 0.6312, 0.5766, 0.5246$ at $c = 1.5, 2, 3$, outside the error bars, whereas the fixed points $d = 0.485, 0.221, 0.061$ give 0.6163, 0.5632, 0.5214. No systematic mismatch was found, so the status "sketch" is limited by the missing concentration proof, not by a numerical discrepancy. The Barak–Erdős phase-transition paper cited by Reisach et al. (2026) concerns monotone-path reachability at the dense scale $\log n/n$ and is not used here.

M. Accuracy Interpretation of s_ρ

An edge $\{x, y\}$ of a DAG is shown in a uniformly random order together with $(\rho_{\text{first}}, \rho_{\text{second}})$. A *pairwise count rule* is $f(a, b) \in [0, 1]$, the probability of declaring "first is the tail"; the *sort rule* is $f_{\text{sort}} = 1, \frac{1}{2}, 0$ for $a <, =, > b$. For the true edge $x \to y$ the accuracy is $\frac{1}{2} f(\rho_x, \rho_y) + \frac{1}{2}(1 - f(\rho_y, \rho_x))$.

Proposition 29 (elementary)**.** *On every DAG, (i) the accuracy of f_{sort} is exactly s_ρ; (ii) every pairwise count rule has accuracy at most s_ρ; (iii) f_{sort} never errs on an edge with $\rho_x \neq \rho_y$.*

Lean: `accRule_sortRule`, `accRule_sortRule_rho`, `accRule_le_sOf`, `accRule_le_sRho`, `edgeAcc_sortRule_strict`, `rho_not_gt_of_edge`.

Proof. By Reisach et al.'s Theorem 1, $\rho_x \leq \rho_y$ on an edge $x \to y$. If $\rho_x < \rho_y$ the accuracy is at most 1, attained by the sort rule; if $\rho_x = \rho_y = a$ it is $\frac{1}{2} f(a, a) + \frac{1}{2}(1 - f(a, a)) = \frac{1}{2}$ for every f, because the random order forces the tie decision to $1/2$. These are the edge scores. (i) restates the definition of s_ρ; (ii) is a three-line consequence of their Theorem 1. □

Combined with Theorem 5, the best accuracy of any pairwise count rule on the stationary unrolled DAG is $\frac{1}{2} + |E_\uparrow|/(2|E_{\text{S}}|)$, at most 0.665 for ER summary graphs and 0.75 for river-like in-forests ($\mu \leq 1$). The restriction to counts is necessary (exhaustive enumeration over labelled DAGs, not proved):

nodes p	2	3	4	5
Bayes rule on the two counts (= edge-pooled s_ρ)	0.5	0.5625	0.592262	0.605547
Bayes rule on counts and the two node degrees	0.5	0.625	0.660714	0.669238

For the skeleton $a - c - b$ the collider has counts $(2, 2, 3)$ (all edges strict) while the chain and the fork have all counts equal to 3; among the tied graphs the centre c is the tail with probability $2/3$ under a uniform prior, so a degree-aware rule beats the coin. So s_ρ is the Bayes accuracy of pairwise count rules only. This section is interpretive, not a new theorem.